

Vellore Institute of Technology, Chennai
BECE407P - ASIC Design

Lab Report: Single-Cycle CPU FSM Control Unit

Name: Shwetank Shekhar

Roll Number: 22BEC1204

Date: 28th October 2025

1. Aim

Create the synthesizable Verilog-based RTL to design a simple Control Unit (Finite State Machine) in Verilog for a single-cycle CPU executing instructions like LOAD, STORE, ADD, and BRANCH. Include instruction decode logic and a logic for control signals of how they derive for register file, ALU, and memory. Show the functional verification, code coverage and perform the logic equivalence check. Implement the physical design (to extract the output stream file .gds) using GPDK 90nm.

2. EDA Tools Used

- **Simulation (Functional Verification):** Cadence Xcelium
 - **Code Coverage:** Cadence Nclaunch & Incisive Metric Center (IMC)
 - **Synthesis & DFT:** Cadence Genus
 - **Logic Equivalence Check (LEC):** Cadence Conformal
 - **ATPG:** Cadence Modus
 - **Physical Design (Place & Route):** Cadence Innovus
-

3. Detailed description of the Designs

The design implements a single-cycle CPU control unit as a Finite State Machine (FSM). A single-cycle CPU's control unit is purely combinational, whereas this single-cycle design uses sequential state registers, which is required for a sequential FSM implementation.

Key Components:

- **FSM States:** The FSM uses 9 states to manage the single-cycle operations: S_IDLE, S_DECODE, S_EXEC_R, S_WB_R, S_EXEC_MEM, S_MEM_READ, S_WB_MEM, S_MEM_WRITE, and S_EXEC_BR.
- **State Registers:** The FSM is implemented using a state register and a next_state register, both 4 bits wide.
- **State Transition Logic:** A sequential always @(posedge clk or negedge rst_n) block

manages the transition from state to next_state and includes an active-low asynchronous reset to S_IDLE.

- **Next State Logic (Combinational):** A combinational always @(*) block implements the state transition logic using a case (state) statement. In the S_DECODE state, a nested case (opcode) block reads the opcode input to branch to the correct execution path (e.g., OP_LOAD transitions to S_EXEC_MEM).
- **Output Logic (Combinational):** A second combinational always @(*) block generates the datapath control signals (RegWrite, ALUSrc, MemRead, etc.). This is a **Moore FSM** implementation, where outputs depend only on the current state. All control signals are set to a default value of 0 at the top of the block, and then asserted to 1 only in the specific states where they are needed (e.g., RegWrite is set to 1 in state S_WB_R).

4. Procedure

The full RTL-to-GDSII flow was executed.

Step 4.1: Functional Verification (Xcelium)

The RTL was verified to ensure correct state transitions and control signal generation for each instruction.

- **Command:** `xrun control_unit_fsm.v control_unit_fsm_tb.v -access +rwc -sv`
- **Result:** The simulation passed. The testbench (control_unit_fsm_tb.v) includes tasks (check_state_and_signals) that step through each clock cycle for all four instruction types and NOP, verifying the FSM state and all 7 control signals at each step.

Step 4.2: Code Coverage (Nclaunch & IMC)

Code coverage was run to ensure the testbench provided adequate stimulus.

- **Command:** `nclaunch` (to open the GUI)
- **Result:** The testbench was designed to be thorough, stepping through every defined instruction path. This exercised all FSM states and transitions, resulting in high block, toggle, and expression coverage in the Incisive Metric Center (IMC).

Step 4.3: Synthesis & DFT (Genus)

The verified RTL was synthesized into a gate-level netlist with scan chains.

- **Command:** `genus -f control_unit_fsm_script_dft.tcl`
- **Process:**
 1. The script read the 90nm slow.lib, the RTL (control_unit_fsm.v), and the timing constraints (control_unit_fsm_sdc.g).

2. DFT was configured for muxed_scan style with a SE (Scan Enable) port.
3. The design was synthesized (syn_generic, syn_map, syn_opt).
4. A single scan chain (top_chain) was defined and connected to the FSM's state registers.
5. The final DFT-inserted netlist (control_unit_fsm_generated_netlist_dft.v) and ATPG files were generated.

Step 4.4: Logic Equivalence Check (Conformal)

LEC was used to mathematically prove that the synthesized netlist was logically identical to the original RTL.

- **Command:** lec -XI -nogui -color -64 -dofile control_unit_fsm_lec.do
- **Process:** The control_unit_fsm_lec.do script was configured to read the golden (RTL) and revised (netlist) designs. DFT pins (scan_in, scan_out, SE) were correctly ignored or constrained to 0.
- **Result:** The comparison passed, reporting **Equivalent**.

Step 4.5: ATPG (Modus)

The ATPG files from Genus were used to generate test patterns for the scan chain.

- **Directory:** synthesis/test_scripts/
- **Command:** modus -f control_unit_fsm_modus.tcl
- **Result:** Modus successfully generated test vectors, confirming the testability of the FSM's sequential elements.

Step 4.6: Physical Design (Innovus)

The final netlist was implemented using the Place & Route flow.

- **Command:** innovus
- **Process:**
 1. **Import:** The DFT netlist and SDC files were imported. MMMC was configured using slow.lib, fast.lib, GPDK 90nm LEF files, and the captable.
 2. **Floorplan:** A floorplan with 0.7 core utilization and an aspect ratio of 1 was defined.
 3. **Power:** Power rings and stripes were created for vdd and vss.
 4. **Placement:** Standard cells were placed.
 5. **CTS:** The clock tree was synthesized.
 6. **Routing:** NanoRoute was used for global and detailed routing.
 7. **Post-Route:** Filler cells were added.
 8. **Export:** The final layout was saved as control_unit_fsm.gds.

5. RTL Codes

Design: control_unit_fsm.v

```
`timescale 1ns / 1ps

// single-Cycle CPU Control Unit (FSM)
module control_unit_fsm (
    input wire clk,
    input wire rst_n, // Active-low asynchronous reset

    // Input from Instruction Register
    input wire [3:0] opcode,

    // Control Signals for Datapath
    output reg RegWrite,
    output reg ALUSrc, // 0: RegFile, 1: Immediate
    output reg [1:0] ALUOp, // 00:ADD, 01:SUB, 10:Opcode-defined
    output reg MemRead,
    output reg MemWrite,
    output reg MemtoReg, // 0: ALU_Result, 1: Mem_Data
    output reg Branch
);

// --- State Definitions ---
parameter [3:0]
    S_IDLE    = 4'b0000,
    S_DECODE  = 4'b0001,
    S_EXEC_R  = 4'b0010, // R-type (ADD) execute
    S_WB_R    = 4'b0011, // R-type (ADD) writeback
    S_EXEC_MEM = 4'b0100, // Mem (LW/SW) address calc
    S_MEM_READ = 4'b0101, // Mem (LW) read
    S_WB_MEM  = 4'b0110, // Mem (LW) writeback
    S_MEM_WRITE = 4'b0111, // Mem (SW) write
    S_EXEC_BR  = 4'b1000; // Branch (BEQ) execute

// --- Opcode Definitions ---
parameter [3:0]
    OP_ADD    = 4'b0000,
    OP_LOAD   = 4'b0001,
    OP_STORE  = 4'b0010,
    OP_BRANCH = 4'b0011;
```

```

// --- State Registers ---
reg [3:0] state, next_state;

// --- State Transition Logic (Sequential) ---
always @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        state <= S_IDLE;
    end else begin
        state <= next_state;
    end
end

// --- Next State Logic (Combinational) ---
always @(*) begin
    case (state)
        S_IDLE:
            next_state = S_DECODE;
        S_DECODE:
            case (opcode)
                OP_ADD: next_state = S_EXEC_R;
                OP_LOAD: next_state = S_EXEC_MEM;
                OP_STORE: next_state = S_EXEC_MEM;
                OP_BRANCH: next_state = S_EXEC_BR;
                default: next_state = S_IDLE; // NOP
            endcase
        S_EXEC_R: next_state = S_WB_R;
        S_WB_R: next_state = S_IDLE;
        S_EXEC_MEM:
            case (opcode)
                OP_LOAD: next_state = S_MEM_READ;
                OP_STORE: next_state = S_MEM_WRITE;
                default: next_state = S_IDLE;
            endcase
        S_MEM_READ: next_state = S_WB_MEM;
        S_WB_MEM: next_state = S_IDLE;
        S_MEM_WRITE: next_state = S_IDLE;
        S_EXEC_BR: next_state = S_IDLE;
        default: next_state = S_IDLE;
    endcase
end

// --- Output Logic (Combinational) ---

```

```

always @(*) begin
    // Default values (all signals 0)
    RegWrite = 1'b0;
    ALUSrc = 1'b0;
    ALUOp = 2'b00;
    MemRead = 1'b0;
    MemWrite = 1'b0;
    MemtoReg = 1'b0;
    Branch = 1'b0;

    case (state)
        S_EXEC_R: begin // ADD
            ALUSrc = 1'b0;
            ALUOp = 2'b10;
        end
        S_WB_R: begin // ADD Writeback
            RegWrite = 1'b1;
            MemtoReg = 1'b0;
        end
        S_EXEC_MEM: begin // LOAD/STORE Address Calc
            ALUSrc = 1'b1;
            ALUOp = 2'b00;
        end
        S_MEM_READ: begin // LOAD
            MemRead = 1'b1;
        end
        S_WB_MEM: begin // LOAD Writeback
            RegWrite = 1'b1;
            MemtoReg = 1'b1;
        end
        S_MEM_WRITE: begin // STORE
            MemWrite = 1'b1;
        end
        S_EXEC_BR: begin // BRANCH
            ALUSrc = 1'b0;
            ALUOp = 2'b01;
            Branch = 1'b1;
        end
    endcase
end
endmodule

```

Testbench: control_unit_fsm_tb.v

```
`timescale 1ns / 1ps

module control_unit_fsm_tb;

    // Parameters
    localparam CLK_PERIOD = 10;

    // Opcode Definitions
    localparam [3:0]
        OP_ADD   = 4'b0000,
        OP_LOAD  = 4'b0001,
        OP_STORE = 4'b0010,
        OP_BRANCH = 4'b0011,
        OP_NOP   = 4'b1111;

    // State Definitions
    localparam [3:0]
        S_IDLE    = 4'b0000,
        S_DECODE  = 4'b0001,
        S_EXEC_R   = 4'b0010,
        S_WB_R    = 4'b0011,
        S_EXEC_MEM = 4'b0100,
        S_MEM_READ = 4'b0101,
        S_WB_MEM  = 4'b0110,
        S_MEM_WRITE = 4'b0111,
        S_EXEC_BR  = 4'b1000;

    // Signals
    reg clk;
    reg rst_n;
    reg [3:0] opcode;

    wire RegWrite, ALUSrc, MemRead, MemWrite, MemtoReg, Branch;
    wire [1:0] ALUOp;

    // Statistics
    integer errors = 0;

    // Instantiate DUT
    control_unit_fsm dut (
        .clk(clk),
```

```

.rst_n(rst_n),
.opcode(opcode),
.RegWrite(RegWrite),
.ALUSrc(ALUSrc),
.ALUOp(ALUOp),
.MemRead(MemRead),
.MemWrite(MemWrite),
.MemtoReg(MemtoReg),
.Branch(Branch)
);

// Clock Generator
initial begin
    clk = 0;
    forever #(CLK_PERIOD / 2) clk = ~clk;
end

// Test Sequence
initial begin
    $dumpfile("wave.vcd");
    $dumpvars(0, dut);

    // 1. Reset Test
    $display("\n--- Test 1: Reset ---");
    rst_n = 0;
    opcode = OP_ADD; // Set an opcode
    #(CLK_PERIOD * 2);
    rst_n = 1;

    @(posedge clk);
    if (dut.state == S_IDLE)
        $display("PASS: FSM reset to S_IDLE.");
    else
        errors = errors + 1;

    // 2. ADD Instruction Test
    $display("\n--- Test 2: ADD Instruction ---");
    opcode = OP_ADD;

    @(posedge clk); // S_IDLE -> S_DECODE
    check_state(S_DECODE);

    @(posedge clk); // S_DECODE -> S_EXEC_R

```

```

check_state_and_signals(S_EXEC_R, 0, 0, 2'b10, 0, 0, 0, 0);

@(posedge clk); // S_EXEC_R -> S_WB_R
check_state_and_signals(S_WB_R, 1, 0, 2'b00, 0, 0, 0, 0);

@(posedge clk); // S_WB_R -> S_IDLE
check_state(S_IDLE);

// 3. LOAD Instruction Test
$display("\n--- Test 3: LOAD Instruction ---");
opcode = OP_LOAD;

@(posedge clk); // S_IDLE -> S_DECODE
check_state(S_DECODE);

@(posedge clk); // S_DECODE -> S_EXEC_MEM
check_state_and_signals(S_EXEC_MEM, 0, 1, 2'b00, 0, 0, 0, 0);

@(posedge clk); // S_EXEC_MEM -> S_MEM_READ
check_state_and_signals(S_MEM_READ, 0, 0, 2'b00, 1, 0, 0, 0);

@(posedge clk); // S_MEM_READ -> S_WB_MEM
check_state_and_signals(S_WB_MEM, 1, 0, 2'b00, 0, 0, 1, 0);

@(posedge clk); // S_WB_MEM -> S_IDLE
check_state(S_IDLE);

// 4. STORE Instruction Test
$display("\n--- Test 4: STORE Instruction ---");
opcode = OP_STORE;

@(posedge clk); // S_IDLE -> S_DECODE
check_state(S_DECODE);

@(posedge clk); // S_DECODE -> S_EXEC_MEM
check_state_and_signals(S_EXEC_MEM, 0, 1, 2'b00, 0, 0, 0, 0);

@(posedge clk); // S_EXEC_MEM -> S_MEM_WRITE
check_state_and_signals(S_MEM_WRITE, 0, 0, 2'b00, 0, 1, 0, 0);

@(posedge clk); // S_MEM_WRITE -> S_IDLE
check_state(S_IDLE);

```

```

// 5. BRANCH Instruction Test
$display("\n--- Test 5: BRANCH Instruction ---");
opcode = OP_BRANCH;

@(posedge clk); // S_IDLE -> S_DECODE
check_state(S_DECODE);

@(posedge clk); // S_DECODE -> S_EXEC_BR
check_state_and_signals(S_EXEC_BR, 0, 0, 2'b01, 0, 0, 0, 1);

@(posedge clk); // S_EXEC_BR -> S_IDLE
check_state(S_IDLE);

// 6. NOP (Default Opcode) Test
$display("\n--- Test 6: NOP (Default) Test ---");
opcode = OP_NOP;

@(posedge clk); // S_IDLE -> S_DECODE
check_state(S_DECODE);

@(posedge clk); // S_DECODE -> S_IDLE
check_state_and_signals(S_IDLE, 0, 0, 2'b00, 0, 0, 0, 0);

// --- Final Result ---
if (errors == 0)
    $display("\n*** ALL TESTS PASSED ***");
else
    $display("\n*** %0d TESTS FAILED ***", errors);

$finish;
end

// Task to check state
task check_state(input [3:0] expected_state);
    if (dut.state == expected_state)
        $display(" PASS: Entered state %h", expected_state);
    else begin
        $display(" FAIL: Expected state %h, but was in %h", expected_state, dut.state);
        errors = errors + 1;
    end
endtask

// Task to check state and all signals

```

```

task check_state_and_signals(
    input [3:0] expected_state,
    input exp_RegWrite,
    input exp_ALUSrc,
    input [1:0] exp_ALUOp,
    input exp_MemRead,
    input exp_MemWrite,
    input exp_MemtoReg,
    input exp_Branch
);
    check_state(expected_state);
    if (RegWrite !== exp_RegWrite) begin $display(" FAIL: RegWrite mismatch"); errors =
errors + 1; end
    if (ALUSrc !== exp_ALUSrc) begin $display(" FAIL: ALUSrc mismatch"); errors = errors
+ 1; end
    if (ALUOp !== exp_ALUOp) begin $display(" FAIL: ALUOp mismatch"); errors = errors
+ 1; end
    if (MemRead !== exp_MemRead) begin $display(" FAIL: MemRead mismatch"); errors =
errors + 1; end
    if (MemWrite !== exp_MemWrite) begin $display(" FAIL: MemWrite mismatch"); errors =
errors + 1; end
    if (MemtoReg !== exp_MemtoReg) begin $display(" FAIL: MemtoReg mismatch"); errors =
errors + 1; end
    if (Branch !== exp_Branch) begin $display(" FAIL: Branch mismatch"); errors = errors
+ 1; end
endtask

endmodule

```

SDC: control_unit_fsm_sdc.g

```

create_clock -name clk -period 10 -waveform {0 5} [get_ports "clk"]
set_clock_transition -rise 0.1 [get_clocks "clk"]
set_clock_transition -fall 0.1 [get_clocks "clk"]
set_clock_uncertainty 0.01 [get_ports "clk"]

# Constrain all inputs
set_input_delay -max 1.0 [get_ports "rst_n"] -clock [get_clocks "clk"]
set_input_delay -max 1.0 [get_ports "opcode"] -clock [get_clocks "clk"]

# Constrain all outputs
set_output_delay -max 1.0 [get_ports "RegWrite"] -clock [get_clocks "clk"]
set_output_delay -max 1.0 [get_ports "ALUSrc"] -clock [get_clocks "clk"]

```

```
set_output_delay -max 1.0 [get_ports "ALUOp"] -clock [get_clocks "clk"]
set_output_delay -max 1.0 [get_ports "MemRead"] -clock [get_clocks "clk"]
set_output_delay -max 1.0 [get_ports "MemWrite"] -clock [get_clocks "clk"]
set_output_delay -max 1.0 [get_ports "MemtoReg"] -clock [get_clocks "clk"]
set_output_delay -max 1.0 [get_ports "Branch"] -clock [get_clocks "clk"]
```

Genus Script: control_unit_fsm_script_dft.tcl

```
set_db init_lib_search_path /home/install/FOUNDRY/digital/90nm/dig/lib
set_db init_hdl_search_path /home/student/Desktop/22BEC1204_ASIC_DESIGN/work
read_libs slow.lib
read_hdl control_unit_fsm.v
elaborate
read_sdc control_unit_fsm_sdc.g

set_db dft_scan_style muxed_scan
set_db dft_prefix dft_
define_shift_enable -name SE -active high -create_port SE
check_dft_rules

set_db syn_generic_effort medium
set_db syn_map_effort medium
set_db syn_opt_effort medium
syn_generic
syn_map
syn_opt

check_dft_rules
set_db design:control_unit_fsm.dft_min_number_of_scan_chains 1
define_scan_chain -name top_chain -sdi scan_in -sdo scan_out -create_ports
connect_scan_chains -auto_create_chains
syn_opt -incr

report_scan_chains
write_dft_atpg -library /home/install/FOUNDRY/digital/90nm/dig/lib

write_hdl > control_unit_fsm_generated_netlist_dft.v
write_sdf -timescale ns -nonegchecks -recrem split -edges check_edge -setuphold split >
control_unit_fsm_generated_delays_dft.sdf
write_sdc > control_unit_fsm_generated_sdc_dft.sdc
write_scandef > control_unit_fsm_generated_scanDEF.scandef

gui_show
```

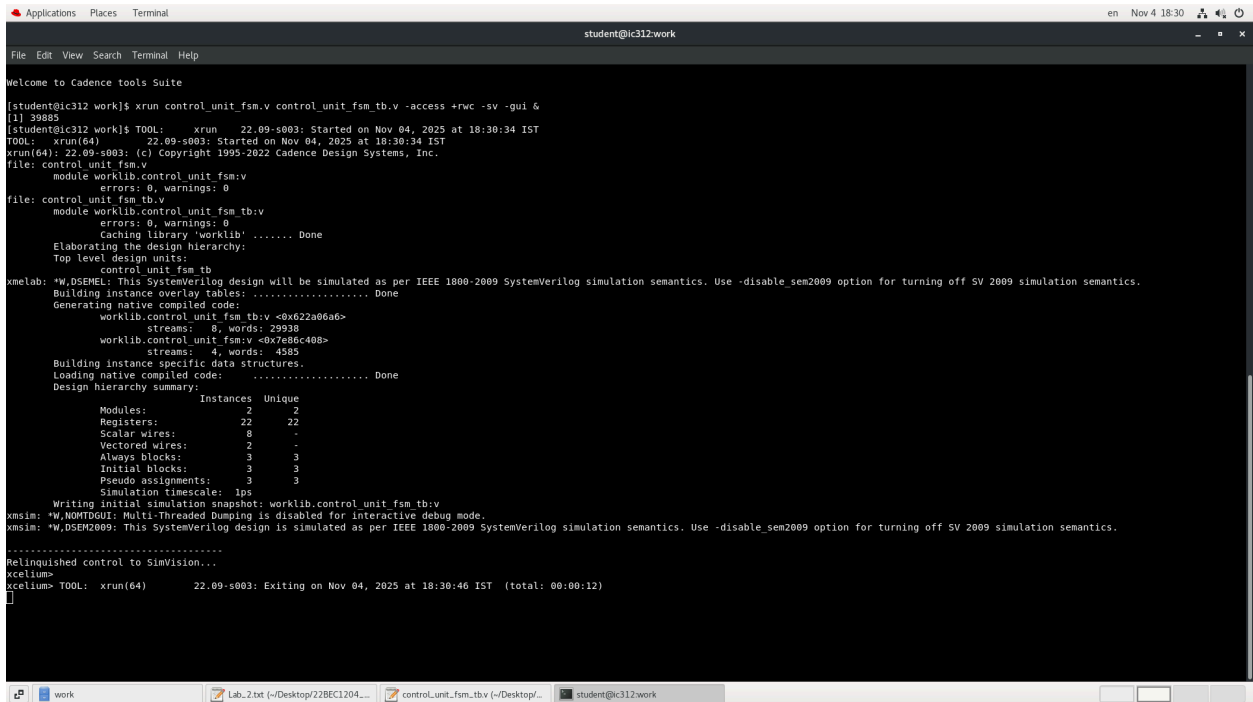
LEC Script: control_unit_fsm_lec.do

```
set log file control_unit_fsm_generated_LEC.log -replace
read library /home/student/22BEC1204_ASIC_DESIGN/LEC/slowlib_verilog_generated.v
-verilog -both
read design /home/student/22BEC1204_ASIC_DESIGN/work/control_unit_fsm.v -verilog
-golden
read design
/home/student/22BEC1204_ASIC_DESIGN/synthesis/control_unit_fsm_generated_netlist_dft.v
-verilog -revised
add ignored inputs scan_in -revised
add ignored outputs scan_out -revised
add pin constraints O SE -revised
set system mode lec
add compare point -all
compare
report verification
set gui on
```

Modus Script: control_unit_fsm_modus.tcl

```
build_model -workdir /home/student/22BEC1204_ASIC_DESIGN/synthesis/test_scripts
-designsource control_unit_fsm.test_netlist.v -techlib
/home/student/22BEC1204_ASIC_DESIGN/LEC/slowlib_verilog_generated.v -designtop
control_unit_fsm
build_testmode -workdir /home/student/22BEC1204_ASIC_DESIGN/synthesis/test_scripts
-testmode FULLSCAN -assignfile control_unit_fsm.FULLSCAN.pinassign
verify_test_structures -workdir
/home/student/22BEC1204_ASIC_DESIGN/synthesis/test_scripts -testmode FULLSCAN
report_test_structures -workdir
/home/student/22BEC1204_ASIC_DESIGN/synthesis/test_scripts -testmode FULLSCAN
build_faultmodel -workdir /home/student/22BEC1204_ASIC_DESIGN/synthesis/test_scripts
-fullfault yes
create_schain_tests -workdir
/home/student/22BEC1204_ASIC_DESIGN/synthesis/test_scripts -testmode FULLSCAN
-experiment scan
create_logic_tests -workdir /home/student/22BEC1204_ASIC_DESIGN/synthesis/test_scripts
-testmode FULLSCAN -experiment logic -effort high
write_vectors -workdir /home/student/22BEC1204_ASIC_DESIGN/synthesis/test_scripts
-testmode FULLSCAN -inexperiment logic -language verilog -scanformat serial
-outputfilename control_unit_fsm_modus_generated_results
```

6. Observations (Waveforms & Screenshots)

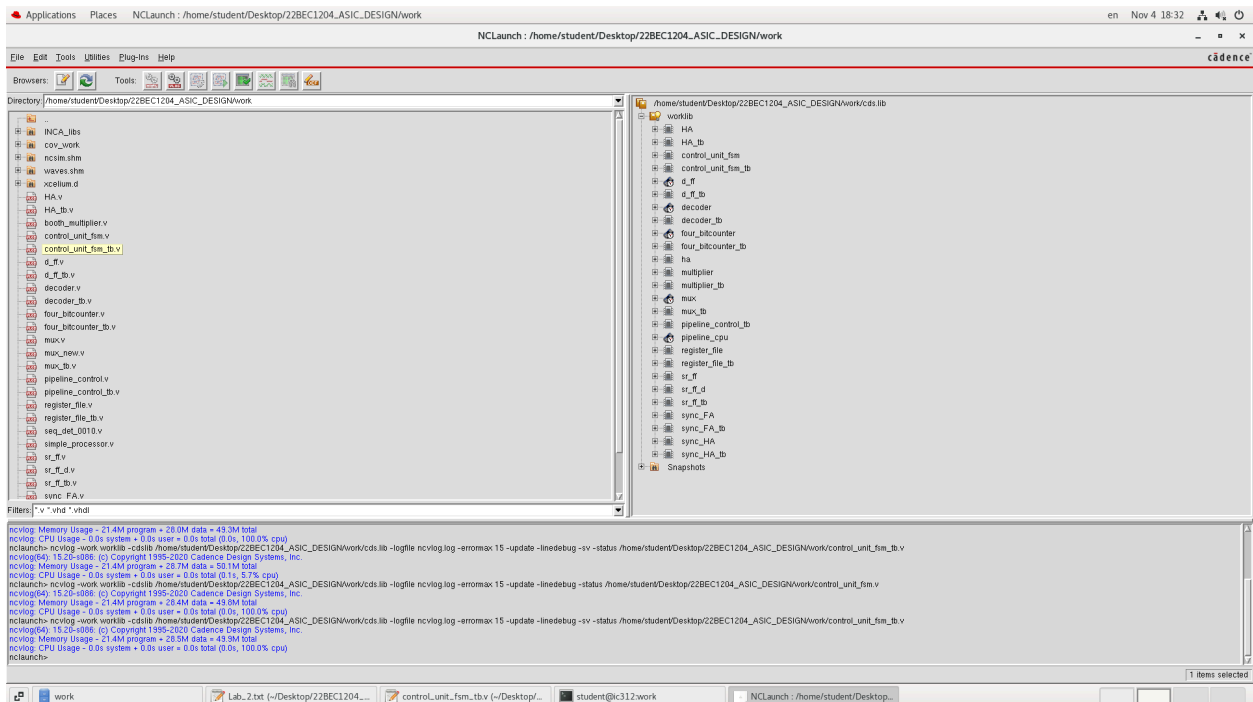


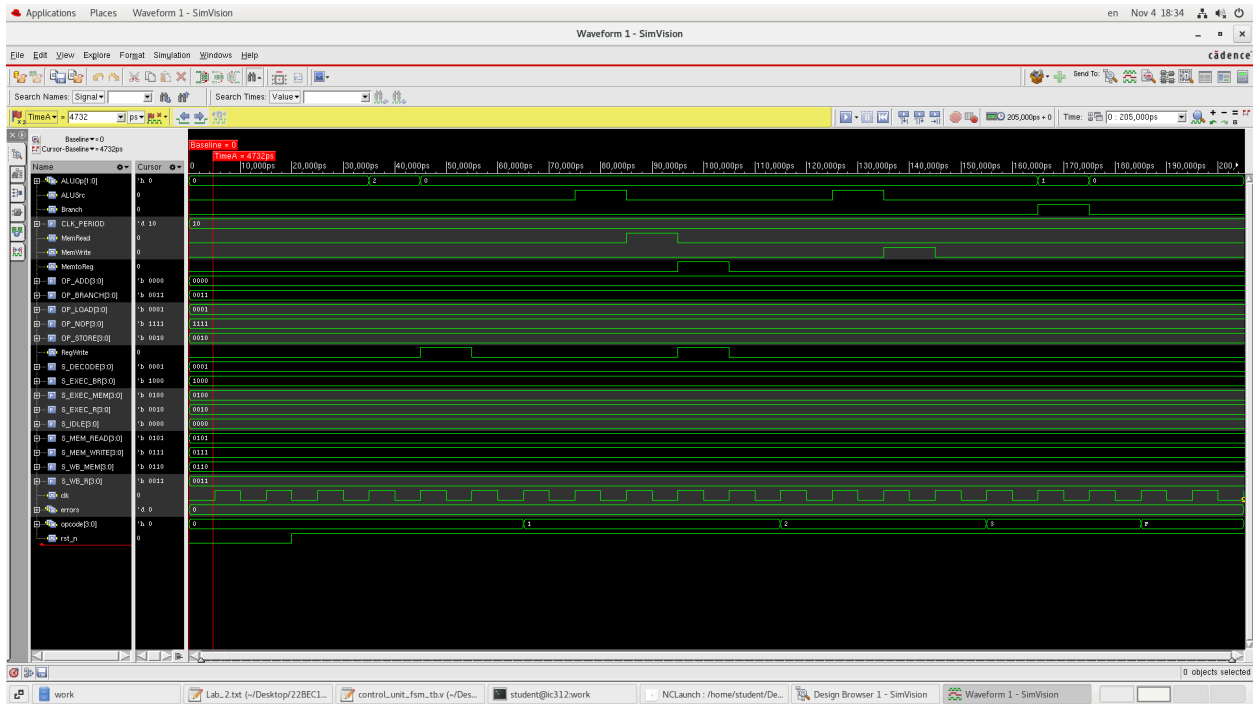
```
Applications  Places  Terminal
student@ic312:work

File Edit View Search Terminal Help

Welcome to Cadence tools Suite

[student@ic312 work]$ xrun control_unit_fsm.v control_unit_fsm.tb.v -access +rwc -sv -gui &
[1] 39885
[student@ic312 work]$ TOOL: xrun 22.09-s003: Started on Nov 04, 2025 at 18:30:34 IST
TOOL: xrun(64): 22.09-s003: Started on Nov 04, 2025 at 18:30:34 IST
xrun(64): 22.09-s003: (c) Copyright 1995-2022 Cadence Design Systems, Inc.
file: control_unit_fsm.v
module worklib.control_unit_fsm:v
errors: 0, warnings: 0
file: control_unit_fsm.tb.v
module worklib.control_unit_fsm.tb:v
errors: 0, warnings: 0
Caching library 'worklib' ..... Done
Elaborating the design hierarchy:
Top level design units:
control_unit_fsm.tb
xcelab: *W, DSEML: This SystemVerilog design will be simulated as per IEEE 1800-2009 SystemVerilog simulation semantics. Use -disable_sem2009 option for turning off SV 2009 simulation semantics.
Building instance overlay tables: ..... Done
Generating native compiled code:
worklib.control_unit_fsm.tb:v <0x622a86a6>
streams: 8, words: 29938
worklib.control_unit_fsm:v <0x7e86c408>
streams: 4, words: 4555
Building instance specific data structures.
Loading native compiled code: ..... Done
Design hierarchy summary:
Instances Unique
Modules: 2 2
Registers: 22 22
Scalar wires: 8 -
Vectored wires: 2 -
Always blocks: 3 3
Initial blocks: 3 3
Pseudo assignments: 3 3
Simulation timescale: lps
Writing initial simulation snapshot: worklib.control_unit_fsm.tb:v
xmsim: *W, NMDT001: Multi-Threaded Dumping is disabled for interactive debug mode.
xmsim: *W, DSEH2009: This SystemVerilog design is simulated as per IEEE 1800-2009 SystemVerilog simulation semantics. Use -disable_sem2009 option for turning off SV 2009 simulation semantics.
.....
Relinquished control to SIMVision...
xcelium> TOOL: xrun(64) 22.09-s003: Exiting on Nov 04, 2025 at 18:30:46 IST (total: 00:00:12)
]
```





Applications Places IMC [Analysis]

en Nov 4 18:34

IMC (64) [Analysis - Metrics]

File View Analysis Help

Load Date Context Views Analyze Refinement Report

Verification Metrics

Relative Elements

Name	Overall Average Grade	Overall Covered	Assertion Status Grade
(no filter)	(no filter)	(no filter)	(no filter)
Types	85.05%	129 / 149 (86.58%)	n/a
Instances	85.05%	129 / 149 (86.58%)	n/a

Showing 2 items

Verification Hierarchy

Name	Overall Average Grade	Overall Covered	Assertion Status Grade
(no filter)	(no filter)	(no filter)	(no filter)
Verification Metrics	85.05%	258 / 299 (86.58%)	n/a
Types	85.05%	129 / 149 (86.58%)	n/a
control_unit_fsm_tb	77.24%	59 / 73 (80.82%)	n/a
control_unit_fsm	92.86%	70 / 76 (92.11%)	n/a
Instances	85.05%	129 / 149 (86.58%)	n/a
control_unit_fsm_tb	85.05%	129 / 149 (86.58%)	n/a
dut	92.86%	70 / 76 (92.11%)	n/a

Showing 7 items

Details

Metrics Source Attributes

Name	Overall Average Grade	Overall Covered
Overall	85.05%	258 / 299 (86.58%)
Code	83.27%	234 / 252 (94.92%)
Block	89.91%	158 / 180 (87.78%)
Expression	76.62%	0 / 0 (n/a)
Toggle	76.62%	56 / 72 (77.78%)
FSM	96.43%	44 / 46 (95.65%)
Functional	n/a	0 / 0 (n/a)
Assertion	n/a	0 / 0 (n/a)
CoverGroup	n/a	0 / 0 (n/a)

Showing 9 items

Loaded Run: /zov_workspace/test

work Lab_2.txt (~/Desktop...) control_unit_fsm_tb... student@ic312:work NCLaunch /home/st... Design Browser 1 - S... Console - SimVision Waveform 1 - SimVis... IMC (64) [Analysis - ...]

Applications Places Terminal

en Nov 4 18:35

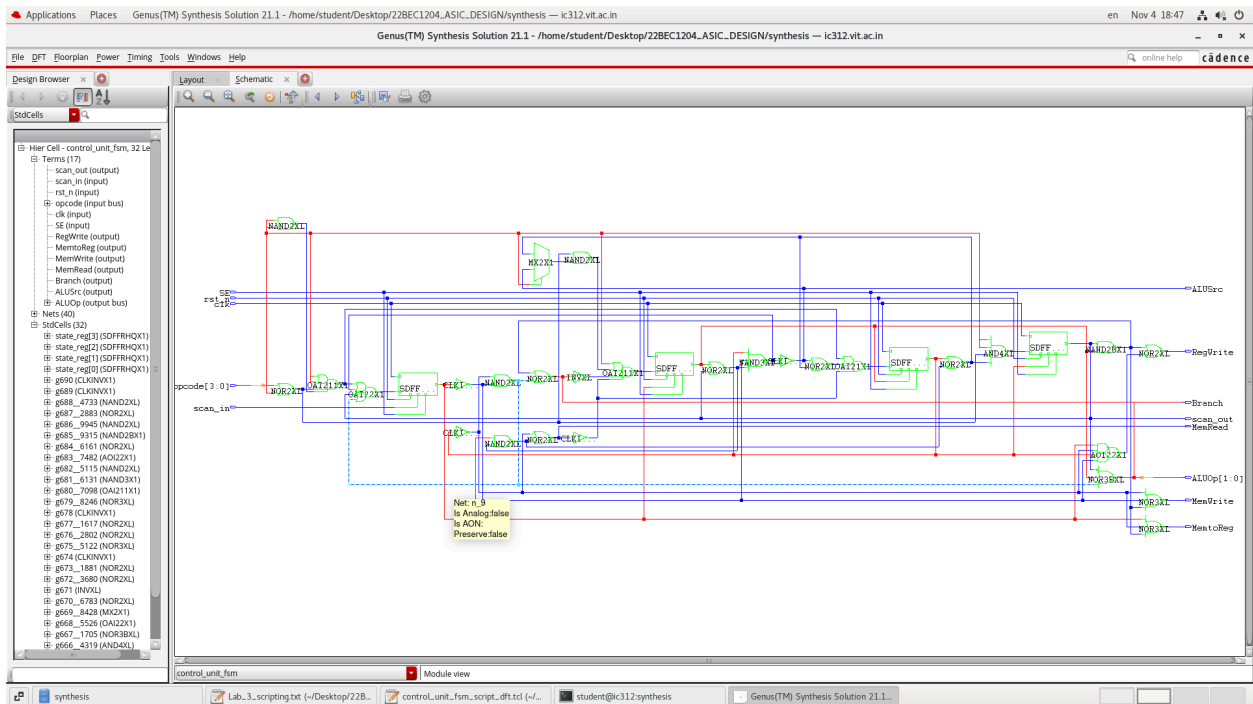
student@ic312:work

File Edit View Search Terminal Help

Welcome to Cadence tools Suite

```
[student@ic312 work]$ xrun control_unit_fsm.v control_unit_fsm_tb.v -access +rwc -sv -gui &
[1] 39885
[student@ic312 work]$ TOOL: xrun 22.09-s003: Started on Nov 04, 2025 at 18:30:34 IST
TOOL: xrun(64): 22.09-s003: Started on Nov 04, 2025 at 18:30:34 IST
xrun(64): 22.09-s003: (c) Copyright 1995-2022 Cadence Design Systems, Inc.
file: control_unit_fsm.v
module worklib.control_unit_fsm.v
errors: 0, warnings: 0
file: control_unit_fsm_tb.v
module worklib.control_unit_fsm_tb.v
errors: 0, warnings: 0
Caching library 'worklib' ..... Done
Elaborating the design hierarchy:
Top level design units:
control_unit_fsm_tb
xmelab: *W, DSEH2009: This SystemVerilog design will be simulated as per IEEE 1800-2009 SystemVerilog simulation semantics. Use -disable_sem2009 option for turning off SV 2009 simulation semantics.
Building instance overlay tables: ..... Done
Generating native compiled code:
worklib.control_unit_fsm_tb.v <0x622a06a6>
streams: 8, words: 29938
worklib.control_unit_fsm.v <0x7e86c408>
streams: 4, words: 4505
Building instance specific data structures.
Loading native compiled code: ..... Done
Design hierarchy summary:
Instances Unique
Modules: 2 2
Registers: 22 22
Scalar wires: 8 -
Vectored wires: 2 -
Always blocks: 3 3
Initial blocks: 3 3
Pseudo assignments: 3 3
Simulation timescale: 1ps
Writing initial simulation snapshot: worklib.control_unit_fsm_tb.v
xmsim: *W, NONTDGUI: Multi-Threaded Dumping is disabled for interactive debug mode.
xmsim: *W, DSEH2009: This SystemVerilog design is simulated as per IEEE 1800-2009 SystemVerilog simulation semantics. Use -disable_sem2009 option for turning off SV 2009 simulation semantics.
.....
Relinquished control to SimVision...
xcelium> TOOL: xrun(64) 22.09-s003: Exiting on Nov 04, 2025 at 18:30:46 IST (total: 00:00:12)
[1] Done xrun control_unit_fsm.v control_unit_fsm_tb.v -access +rwc -sv -gui
[student@ic312 work]$ nclaunch
nclaunch(64): 15.20-s006: (c) Copyright 1995-2020 Cadence Design Systems, Inc.
[student@ic312 work]$
```

work Lab_2.txt (~/Desktop/22BEC1204...) control_unit_fsm_tb.v (~/Desktop/... student@ic312:work



Applications Places Conformal(R) Logic Equivalence Checking Nov 5 16:49

Conformal(R) Logic Equivalence Checking

File Compare Tools Custom Preferences Window Help

Design Library Tool Setup Design Data Rule Checking

Current Mode: LEC

File List (Golden):

/home/student/22BEC1204_ASIC_DESIGN/work/control_unit_fsm.v

File List (Revised):

/home/student/22BEC1204_ASIC_DESIGN/synthesis/control_unit_fsm_generated_netlist.v

Design Data Summary

Design	Golden	Revised
Design-Modules	1	1
Library-Cells	10	32
Primitives		
INPUT	6	8
OUTPUT	8	9
AND	78	13
DFF	4	4
INV	6	16
NAND	0	10
NOR	13	11
OR	51	10
WIRE	0	1
World-Level		
WSEL	10	0
Total	152	65

//Note: Enable Unified Schematic Viewer usage.
//Note: Enable buffer/inverter collapsing usage on schematics
//Note: Enable bus collapsing usage on schematics
//Note: Enable independent UI styles on each schematics
//Note: Set schematic traverse max_level to 200 !!!
//Note: Set schematic traverse max_time to 300(seconds) !!!
//Note: Enable Schematic concise drawing support
CPU time : 0.28 seconds
Blaspe time : 1 seconds
Memory usage : 159.91 M bytes

LEC>

Google Gemini - Google Chrome LEC control_unit_fsm Lec.do (~22BEC1204_ (Lab_4_LEC_guide.txt (~22BEC1204_ student@ic312: LEC Conformal(R) Logic Equivalence Che... 0%

Design Data Summary

Design	Golden	Revised
Design-Modules	1	1
Library-Cells	10	32
Primitives		
INPUT	6	8
OUTPUT	8	9
AND	78	13
DFF	4	4
INV	6	16
NAND	0	10
NOR	13	11
OR	51	10
WIRE	0	1
Word-Level		
WSEL	10	0
Total	152	65

```

Applications  Places  Terminal
Nov 5 16:49
student@ic312:LEC

File Edit View Search Terminal Help
// Modeling Revised ...
// Warning: Golden and Revised have different numbers of key points:
// Golden key points = 18
// Revised key points = 19
// Mapping key points ...
=====
Mapped points: SYSTEM class
=====
Mapped points:  PI  PO  DFF  Total
=====
Golden      6   8   4   18
=====
Revised     6   8   4   18
=====
Unmapped points:
=====
Revised:
=====
Unmapped points:  PI  Total
=====
Extra      1   1
=====
// Command: add compare point -all
// 12 compared points added to compare list
// Command: compare
=====
Compared points:  PO  DFF  Total
=====
Equivalent      8   4   12
=====
// Command: report verification
=====
Verification Report
=====
Category                                     Count
-----
1. Non-standard modeling options used:      0
-----
2. Incomplete verification:                  1
   Command "add ignore outputs" used:      yes
-----
3. User modification to design:              0
-----
4. Conformal Constraint Designer clock domain crossing checks recommended: 0
-----
5. Design ambiguity:                         0
-----
6. Compare Results:                         PASS
=====
// Command: set gui on
// Switching to GUI mode
[student@ic312 LEC]$

```

=====

Verification Report

Category

Count

1. Non-standard modeling options used:

0

2. Incomplete verification:

1

Command "add ignore outputs" used:

yes

3. User modification to design:

0

4. Conformal Constraint Designer clock domain crossing checks recommended:

0

5. Design ambiguity:

0

6. Compare Results:

PASS

=====

// Command: set gui on

// Switching to GUI mode

[student@ic312 LEC]\$

Google Gemini - Google Chrome

LEC

control_unit_fsm_le

Applications Places Terminal

Nov 5 5:33 PM

student@ic312:test_scripts

File Edit View Search Terminal Help

1.2.1.3.8	60.25	0.93	60.25	18.13	0.00	18.13	6	5	72
1.2.1.3.9	63.04	2.80	63.04	18.13	0.00	18.13	6	5	78
1.2.1.3.10	64.60	1.55	64.60	18.13	0.00	18.13	6	5	84
1.2.1.4.1	71.45	6.85	71.45	18.13	0.00	18.13	7	5	91
1.2.1.4.2	73.29	1.86	73.29	18.13	0.00	18.13	7	5	98
1.2.1.4.3	85.09	11.80	85.09	18.13	0.00	18.13	7	5	105
1.2.1.5.1	89.13	4.04	89.13	18.13	0.00	18.13	6	5	111
1.2.1.5.2	90.99	1.86	90.99	18.13	0.00	18.13	6	5	117
1.2.1.5.3	93.17	2.17	93.17	18.13	0.00	18.13	6	5	123
1.2.1.5.4	93.79	0.62	93.79	18.13	0.00	18.13	6	5	129
1.2.1.5.5	96.27	2.40	96.27	18.13	0.00	18.13	6	5	135
1.2.1.5.6	97.83	1.55	97.83	18.13	0.00	18.13	6	5	141
1.2.1.5.7	99.07	1.24	99.07	18.13	0.00	18.13	6	5	147
1.2.1.5.8	99.69	0.62	99.69	18.13	0.00	18.13	6	5	153
1.2.1.5.9	100.00	0.31	100.00	18.13	0.00	18.13	7	0	160

[end TVE_050]

INFO (TDA-001): Maximum Memory used during the run and Cumulative Time in hours:minutes:seconds:
Total Memory = 6,050,880 bytes
CPU Time = 0:00:00.00
Elapsed Time = 0:00:00.40 [end TDA_001]
Date Ended: Wednesday Nov 05 17:33:38 2025 IST

INFO (TVE-002): Verilog write vectors has completed. [end TVE_002]

Message Summary

Count	Number	First Instance of Message Text
1	1	INFO (TVE-001): Verilog write vectors started.
1	1	INFO (TVE-002): Verilog write vectors has completed.
1	1	INFO (TVE-003): Verilog write vectors output file will be: /home/student/22BEC1204_ASIC_DESIGN/synthesis/test_scripts/testresults/verilog/control_unit_fsm_modus_generated_results.cyclemap.
1	1	INFO (TVE-004): Reading test section 1.1. Test section type equals scan.
2	1	INFO (TVE-005): Created 12 total cycles, of which 4 are test cycles, 8 are scan cycles, 0 are dynamic timed cycles and 0 are dynamic cycles that are not timed.
2	1	INFO (TVE-008): Created 12 total measures, of which 0 are PO measures and 4 are SO (Scan Out) measures.
1	1	INFO (TVE-050): TEST SEQUENCE COVERAGE ESTIMATE REPORT
3	1	INFO (TVE-103): There was no specified TEST offset for this clock 'pin clk'. A default clock offset of 8.000000 ns will be used.

For a detailed explanation of a message and a suggested user response execute 'msgHelp <message id>'. For example: msgHelp TDA-009

#@ End verbose source control_unit_fsm_modus.tcl
0
modus:root:/ 2>

verilog

student@ic312:test_scripts

control_unit_fsm_modus.tcl (-v22B...

Applications Places Text Editor

Nov 5 9:34 PM

Open [Icons]

control_unit_fsm_modus_generated_results.cyclemap
~\Z2BEC1204_ASIC_DESIGN\synthesis\test_scripts\testresults\verilog\control_unit_fsm_modus_generated_results.cyclemap

Save [Icons] - [Icons] x

resource_shared_datapath_modus.tcl pipeline_control_mo_ted_results.cyclemap register_file_modus_ted_results.cyclemap control_unit_fsm_mo_ted_results.cyclemap

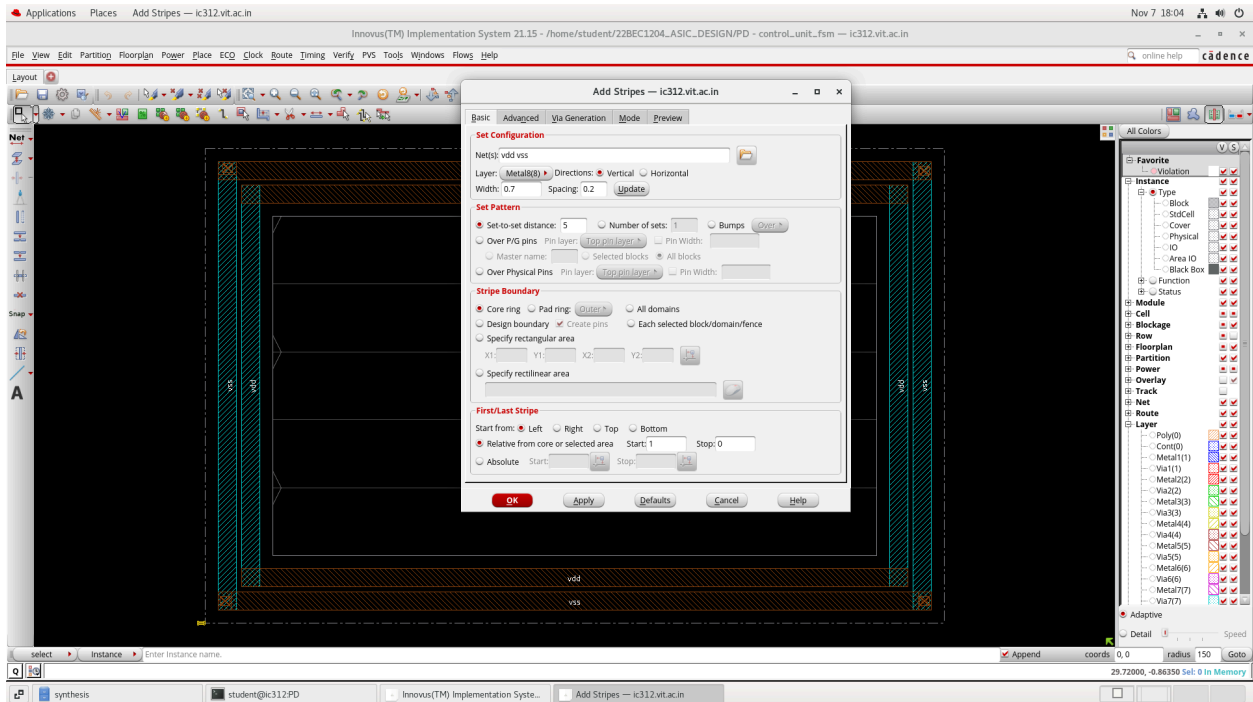
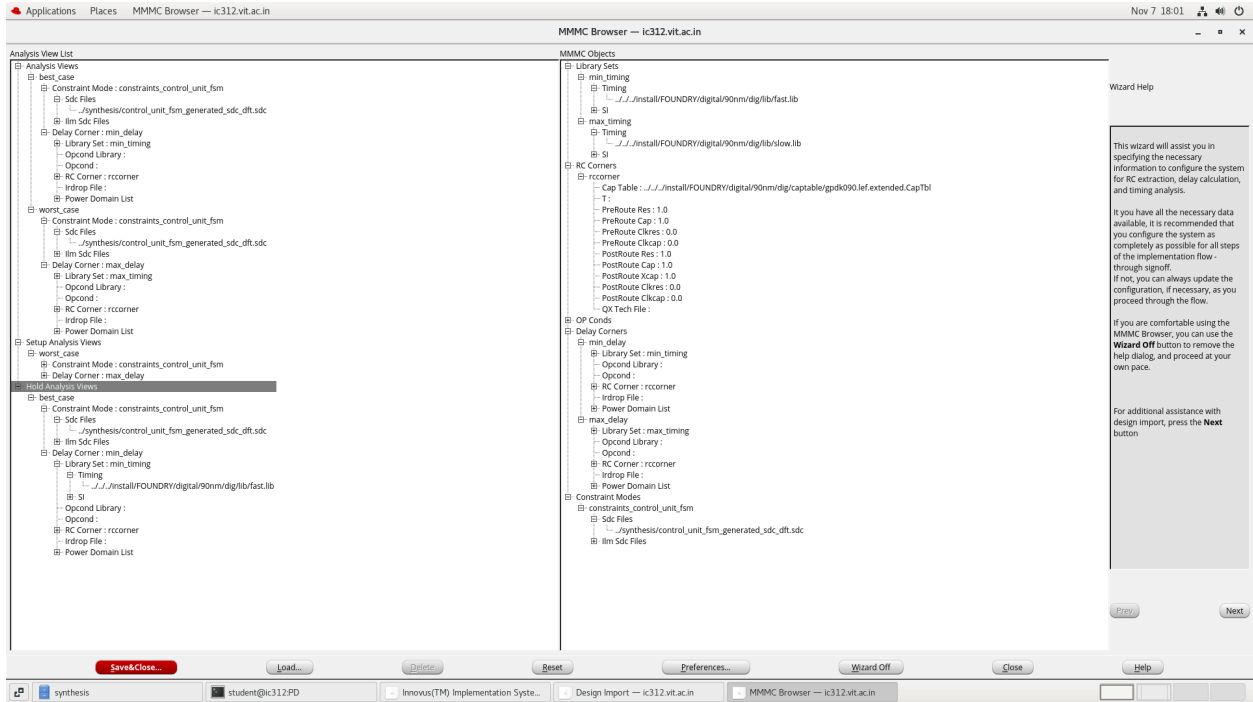
```
// cadence(R) Modus(TM) DFT Software Solution, Version 20.12-s002.1, built Feb 26 2021 (linux26_64)
// This file was created on.....November 05, 2025 at 17:33:30
// The date time stamp for the TDBin file TDBin.FULLSCAN.logic; 20251105120337
//-----
//-----
//-----
// Total      Relative   Relative    Test     CPP       Pattern    Event
// Cycle      Count     Time (ns)  Sequence Number  Odometer   Type
// Count                               Number
//-----
// The vector file output name: control_unit_fsm_modus_generated_results.1.verilog
//-----
//-----
//-----
// The vector file output name: control_unit_fsm_modus_generated_results.2.verilog
//-----
//-----
//-----
```

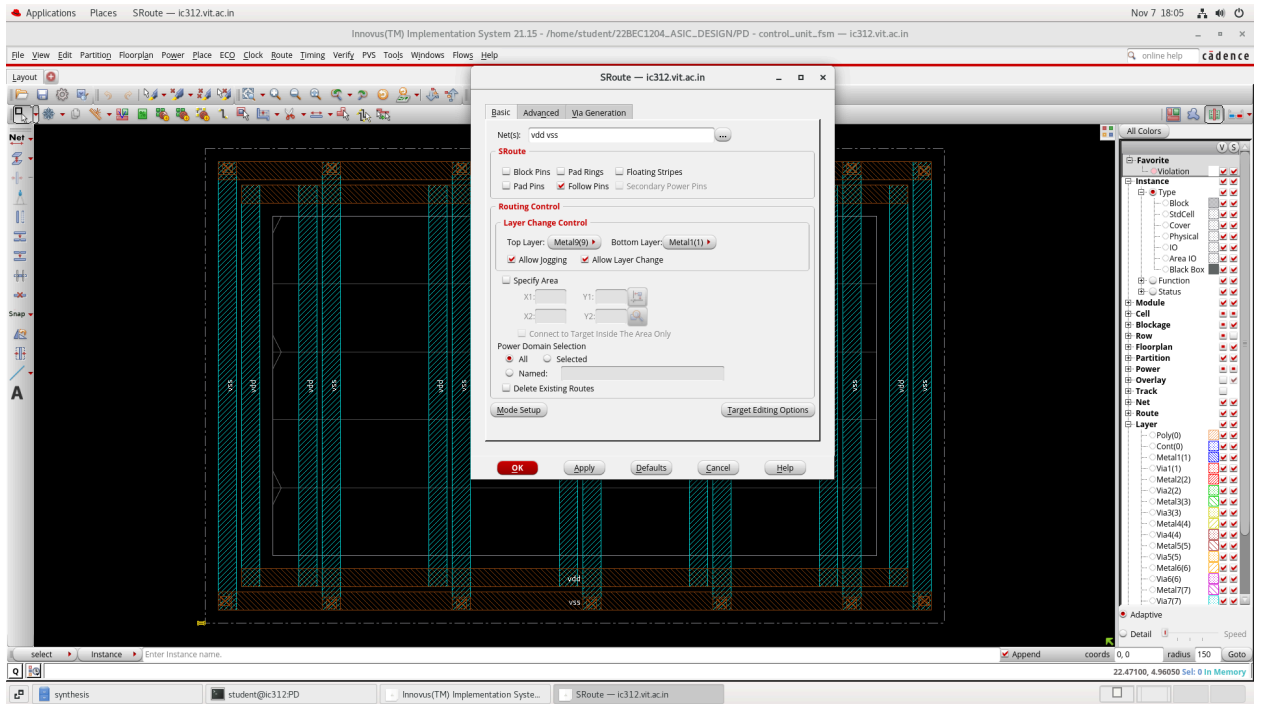
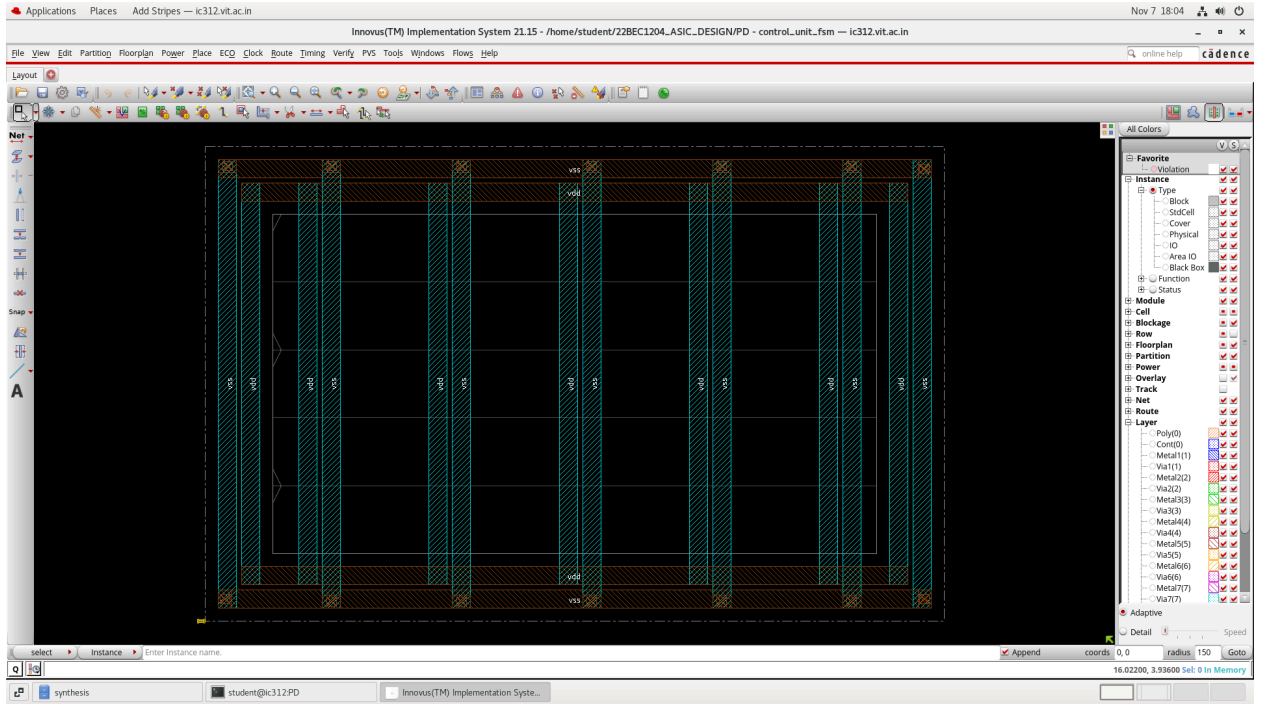
Total Cycle Count	Relative Count	Relative Time (ns)	Test Sequence Number	CPP Sequence Number	Pattern Odometer	Event Type	Cycle Offset	Scan Length	Relative Scan Number	Measure Event
6	6	160	1	2	1.1.1.2.1.1	Scan Load - Shift	0	4	1	N
6	6	480	1	2	1.1.1.2.1.1	end_of Scan_Load	3	0	1	N
8	8	560	2	2	1.1.1.2.1.2	Scan Unload - Shift	0	4	2	N
11	11	880	1	2	1.1.1.2.1.2	end_of Scan_Unload	3	0	2	Y
15	3	160	2	4	1.2.1.3.1.1	Scan Load - Shift	0	4	1	N
18	6	480	2	4	1.2.1.3.1.1	end_of Scan_Load	3	0	1	N
21	9	640	2	4	1.2.1.3.1.4	Scan Unload - Shift - Overlap	0	4	2	Y
21	9	640	3	5	1.2.1.3.1.1	Scan Load - Shift - Overlap	0	4	2	N
24	12	960	3	5	1.2.1.3.1.1	end_of Scan_Load - Overlap	3	0	2	Y
25	13	960	3	5	1.2.1.3.1.3	Measure P0	0	1	2	Y
27	15	1120	5	5	1.2.1.3.1.4	Scan Unload - Shift - Overlap	0	4	3	Y
27	15	1120	4	6	1.2.1.3.2.1	Scan Load - Shift - Overlap	0	4	3	N
30	18	1440	4	6	1.2.1.3.2.1	end_of Scan_Load - Overlap	3	0	3	Y
31	19	1440	4	6	1.2.1.3.2.3	Measure P0	0	1	3	Y
33	21	1600	4	6	1.2.1.3.2.4	Scan Unload - Shift - Overlap	0	4	4	N
33	21	1600	5	7	1.2.1.3.3.1	Scan Load - Shift - Overlap	0	4	4	Y
36	24	1920	5	7	1.2.1.3.3.1	end_of Scan_Load - Overlap	3	0	4	Y
37	25	1920	5	7	1.2.1.3.3.3	Measure P0	0	1	4	Y
39	27	2080	5	7	1.2.1.3.3.4	Scan Unload - Shift - Overlap	0	4	4	N
39	27	2080	6	8	1.2.1.3.4.1	Scan Load - Shift - Overlap	0	4	5	N
42	30	2400	6	8	1.2.1.3.4.1	end_of Scan_Load - Overlap	3	0	5	Y
43	31	2400	6	8	1.2.1.3.4.3	Measure P0	0	1	5	Y
45	33	2560	6	8	1.2.1.3.4.4	Scan Unload - Shift - Overlap	0	4	6	Y
45	33	2560	7	9	1.2.1.3.5.1	Scan Load - Shift - Overlap	0	4	6	N
48	36	2880	7	9	1.2.1.3.5.1	end_of Scan_Load - Overlap	3	0	6	Y
49	37	2880	7	9	1.2.1.3.5.3	Measure P0	0	1	6	Y
51	39	3040	7	9	1.2.1.3.5.4	Scan Unload - Shift - Overlap	0	4	7	Y
51	39	3040	8	10	1.2.1.3.6.1	Scan Load - Shift - Overlap	0	4	7	N
54	42	3360	8	10	1.2.1.3.6.1	end_of Scan_Load - Overlap	3	0	7	Y
55	43	3360	8	10	1.2.1.3.6.3	Measure P0	0	1	7	Y
57	45	3520	8	10	1.2.1.3.6.4	Scan Unload - Shift - Overlap	0	4	8	Y
57	45	3520	9	11	1.2.1.3.7.1	Scan Load - Shift - Overlap	0	4	8	N
60	48	3840	9	11	1.2.1.3.7.1	end_of Scan_Load - Overlap	3	0	8	Y
61	49	3840	9	11	1.2.1.3.7.3	Measure P0	0	1	8	Y
63	51	4000	9	11	1.2.1.3.7.4	Scan Unload - Shift - Overlap	0	4	9	Y
63	51	4000	10	12	1.2.1.3.8.1	Scan Load - Shift - Overlap	0	4	9	N
66	54	4320	10	12	1.2.1.3.8.1	end_of Scan_Load - Overlap	3	0	9	Y

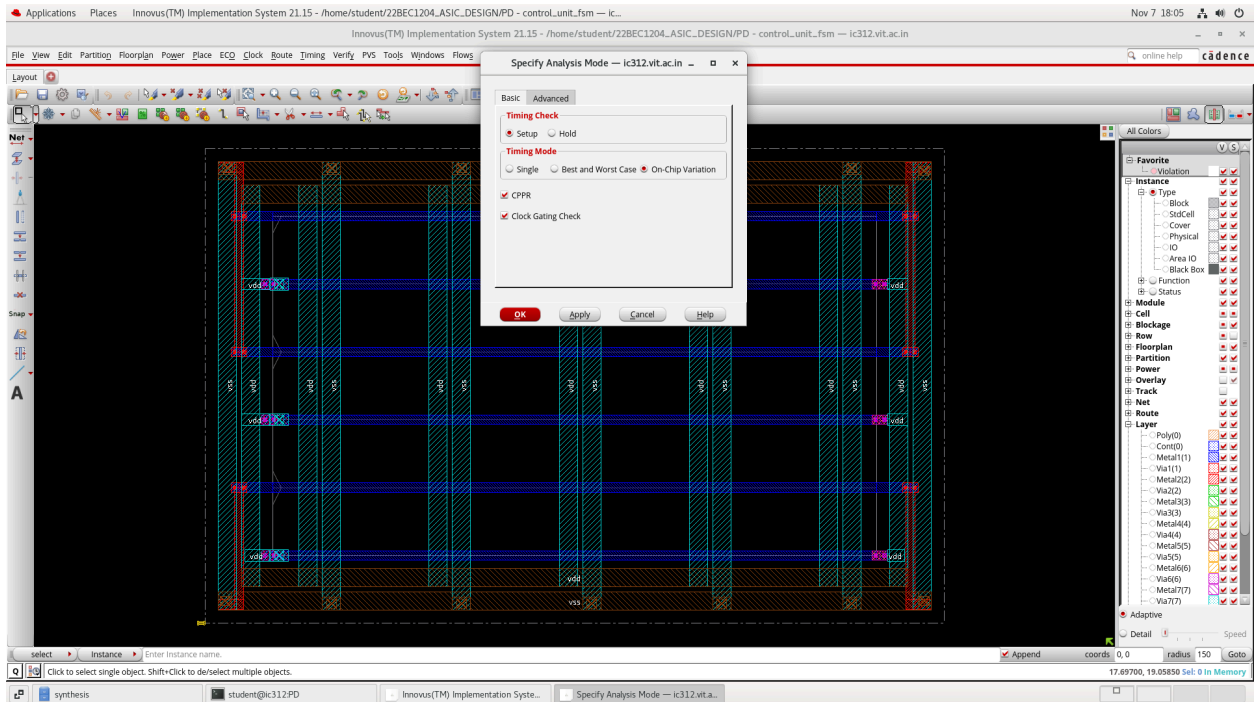
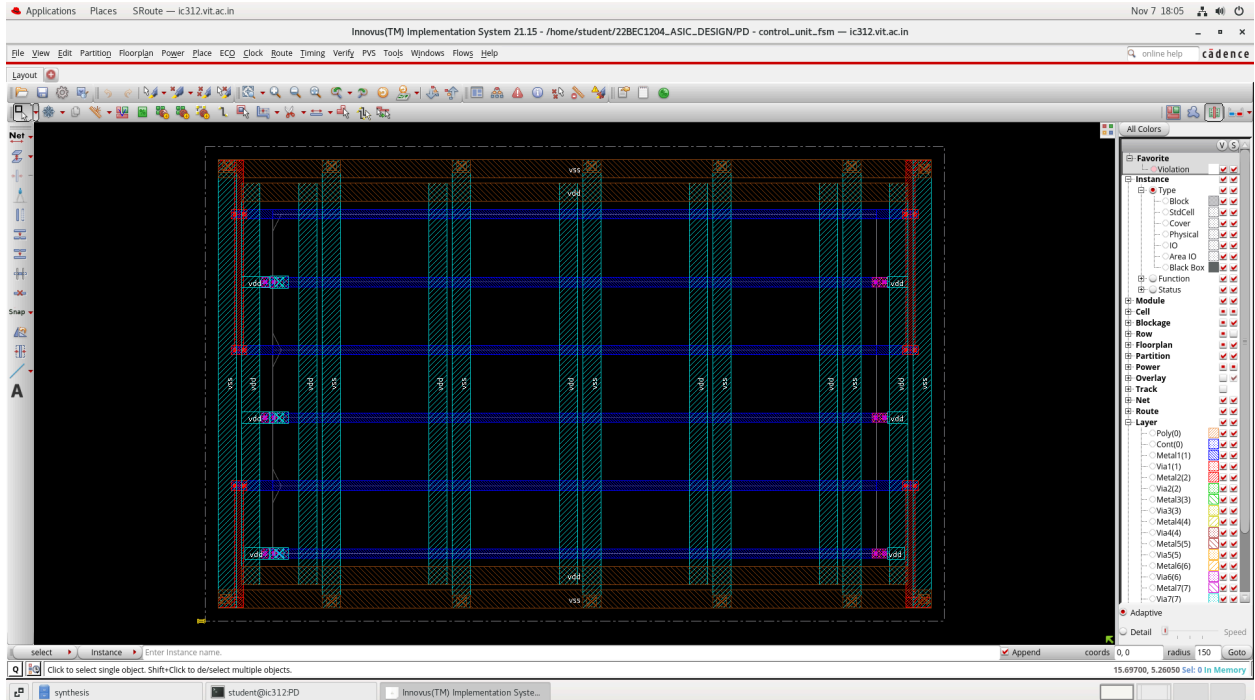
Loading file "home/student/22BEC1204_ASIC_DESIGN/synthesis/test_scripts/testresults/verilog/control_unit_fsm_modus_generated_results.cyclemap"...

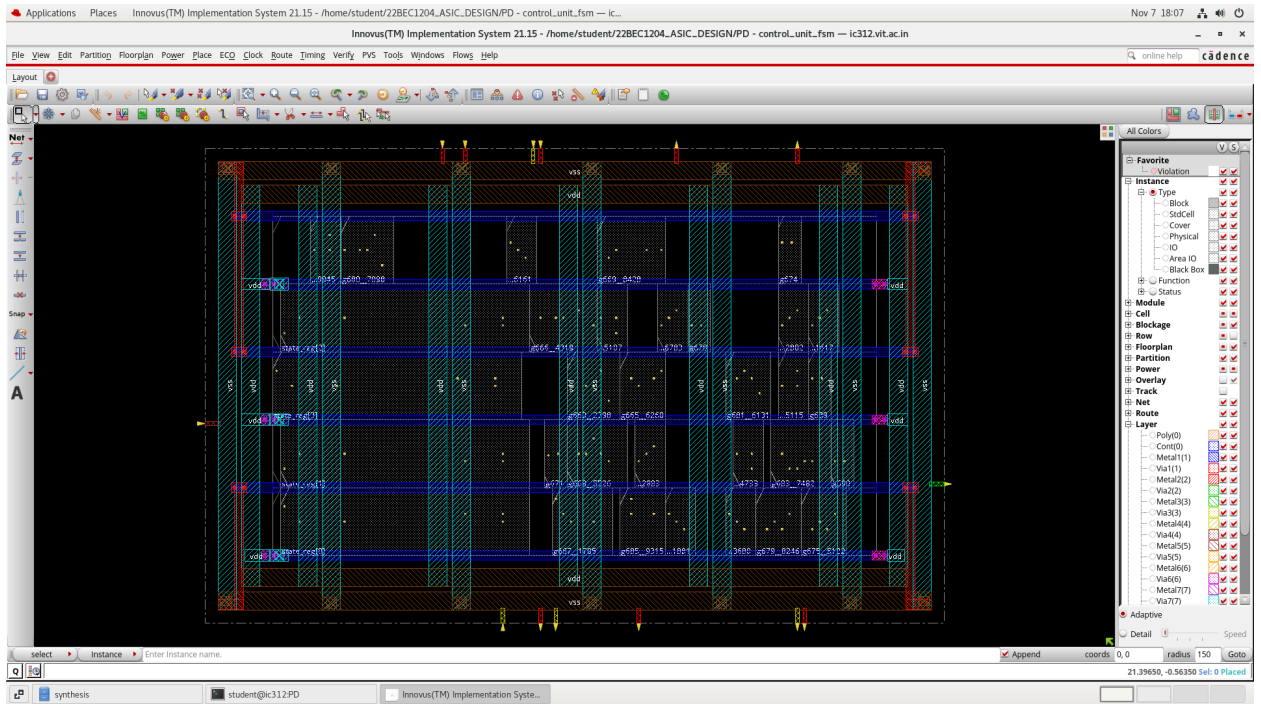
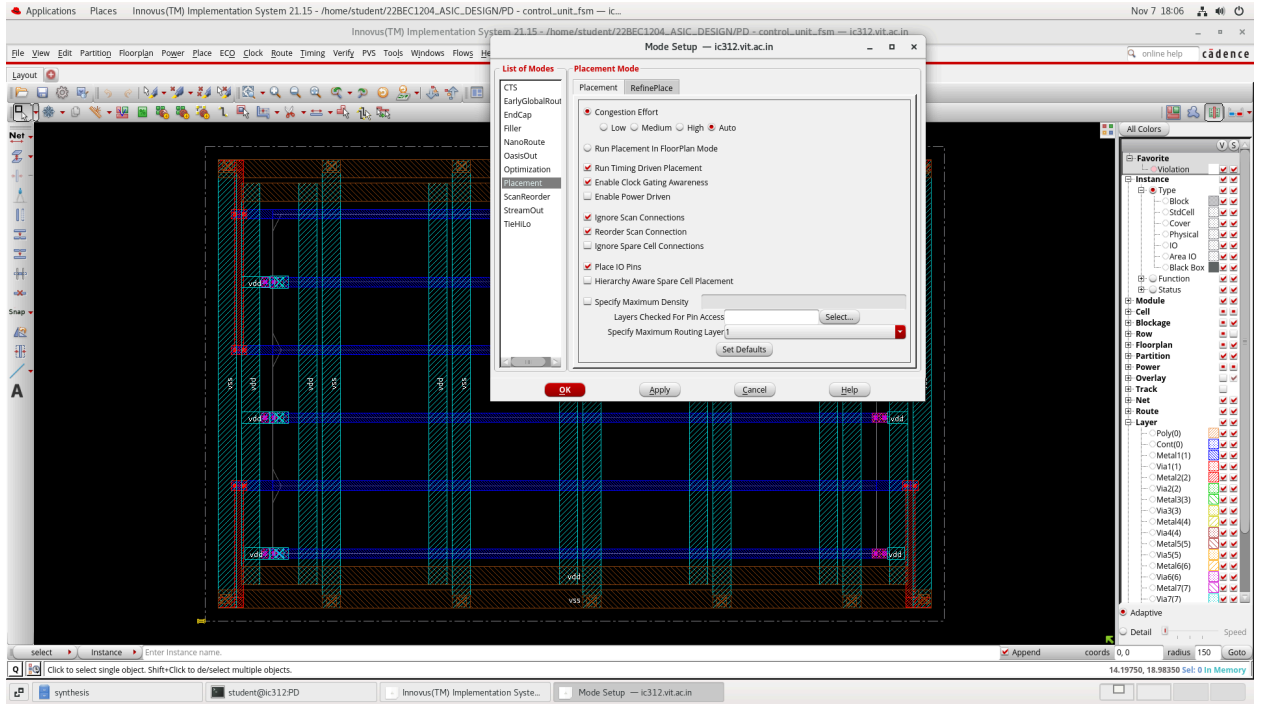
C Tab Width: 8 Ln 1, Col 1 INS

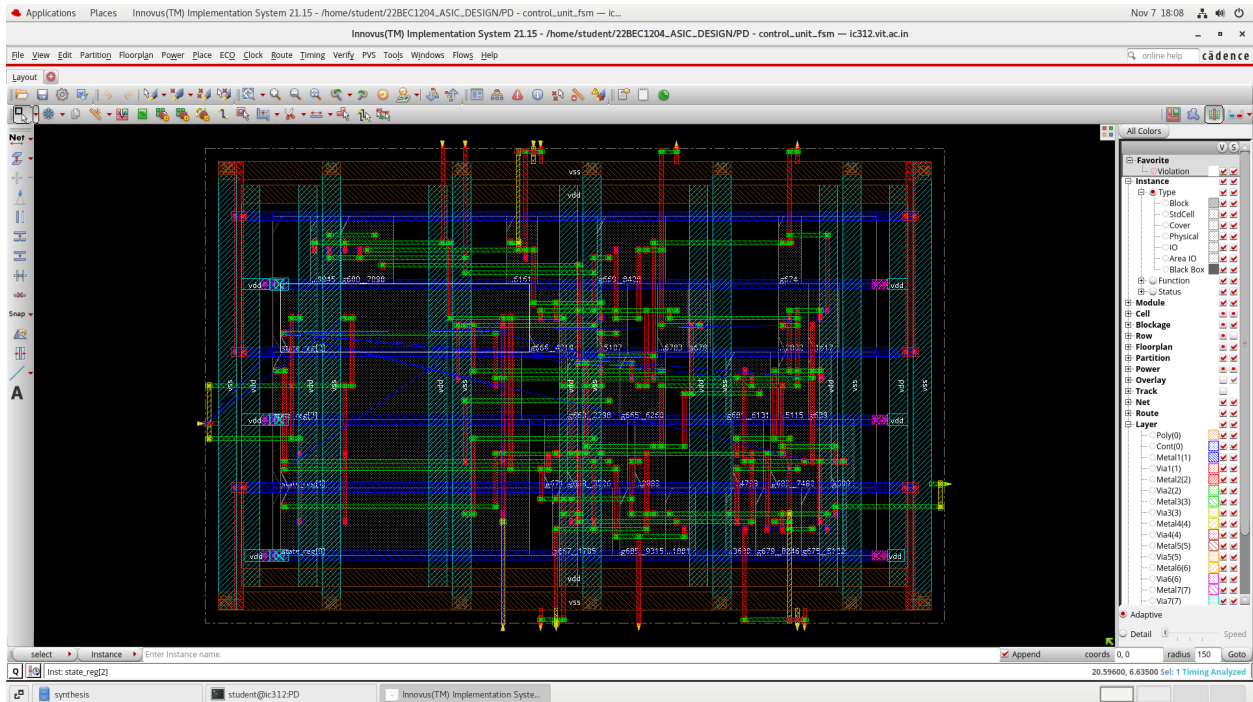
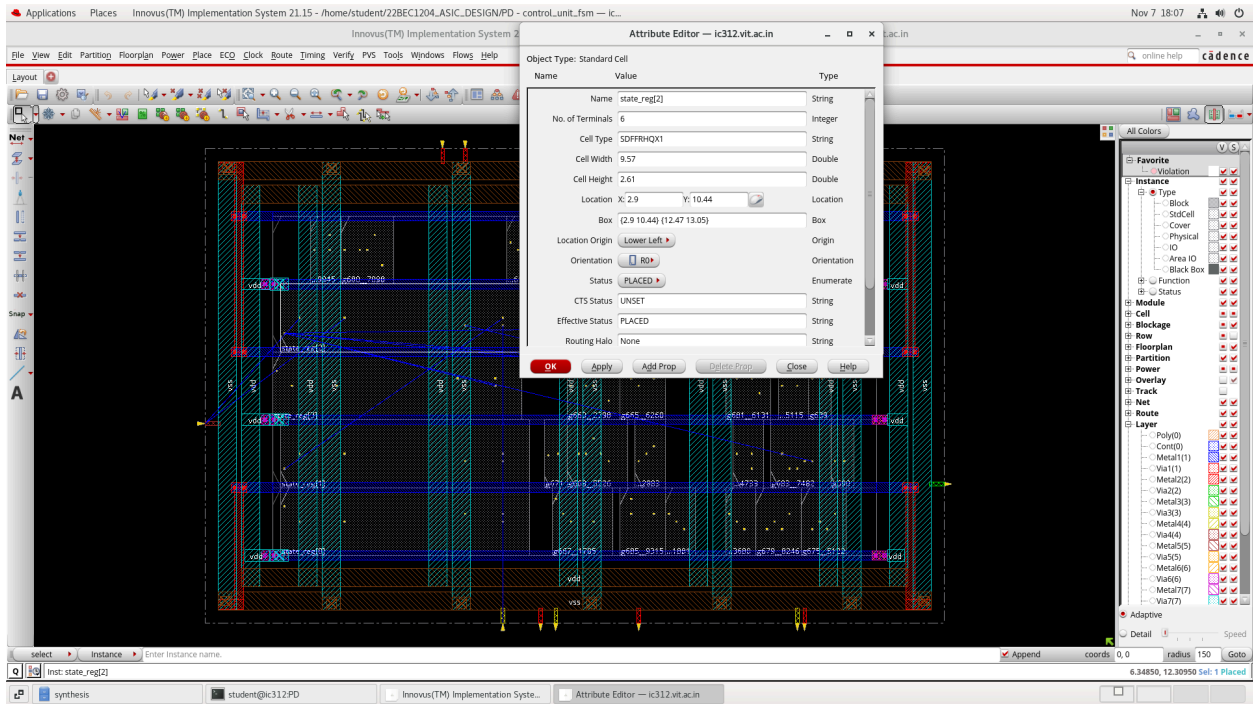
[Icons] verilog student@c312_test_scripts control_unit_fsm_modus_generate...

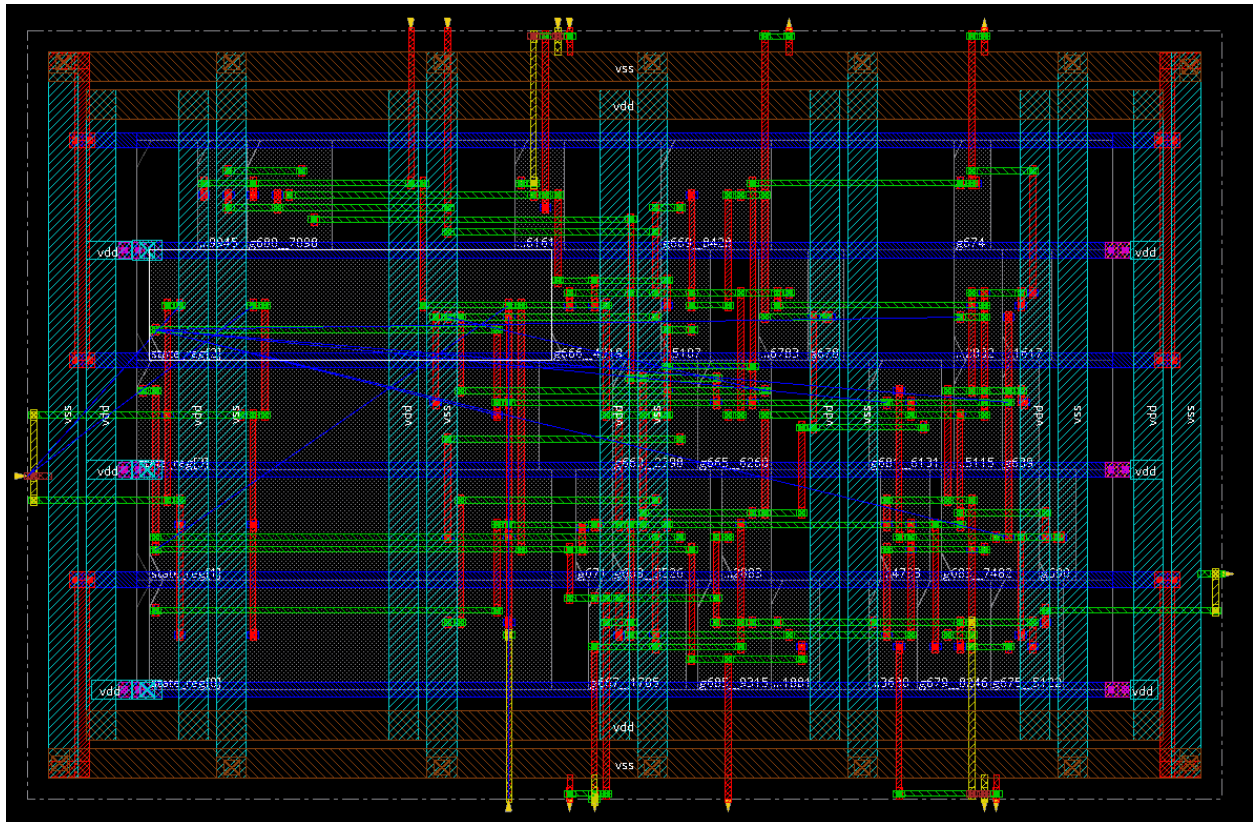












Applications Places Innovus(TM) Implementation System 21.15 - /home/student/22BEC1204_ASIC_DESIGN/DPD - control_unit_fsm - ic...

Innovus(TM) Implementation System 21.15

File View Edit Partition Floorplan Power Place ECO Clock Route Timing Verify PVS Tools Windows Flows Help

Layout

Net

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

21

22

23

24

25

26

27

28

29

30

31

32

33

34

35

36

37

38

39

40

41

42

43

44

45

46

47

48

49

50

51

52

53

54

55

56

57

58

59

60

61

62

63

64

65

66

67

68

69

70

71

72

73

74

75

76

77

78

79

80

81

82

83

84

85

86

87

88

89

90

91

92

93

94

95

96

97

98

99

100

101

102

103

104

105

106

107

108

109

110

111

112

113

114

115

116

117

118

119

120

121

122

123

124

125

126

127

128

129

130

131

132

133

134

135

136

137

138

139

140

141

142

143

144

145

146

147

148

149

150

151

152

153

154

155

156

157

158

159

160

161

162

163

164

165

166

167

168

169

170

171

172

173

174

175

176

177

178

179

180

181

182

183

184

185

186

187

188

189

190

191

192

193

194

195

196

197

198

199

200

201

202

203

204

205

206

207

208

209

210

211

212

213

214

215

216

217

218

219

220

221

222

223

224

225

226

227

228

229

230

231

232

233

234

235

236

237

238

239

240

241

242

243

244

245

246

247

248

249

250

251

252

253

254

255

256

257

258

259

260

261

262

263

264

265

266

267

268

269

270

271

272

273

274

275

276

277

278

279

280

281

282

283

284

285

286

287

288

289

290

291

292

293

294

295

296

297

298

299

300

301

302

303

304

305

306

307

308

309

310

311

312

313

314

315

316

317

318

319

320

321

322

323

324

325

326

327

328

329

330

331

332

333

334

335

336

337

338

339

340

341

342

343

344

345

346

347

348

349

350

351

352

353

354

355

356

357

358

359

360

361

362

363

364

365

366

367

368

369

370

371

372

373

374

375

376

377

378

379

380

381

382

383

384

385

386

387

388

389

390

391

392

393

394

395

396

397

398

399

400

401

402

403

404

405

406

407

408

409

410

411

412

413

414

415

416

417

418

419

420

421

422

423

424

425

426

427

428

429

430

431

432

433

434

435

436

437

438

439

440

441

442

443

444

445

446

447

448

449

450

451

452

453

454

455

456

457

458

459

460

461

462

463

464

465

466

467

468

469

470

471

472

473

474

475

476

477

478

479

480

481

482

483

484

485

486

487

488

489

490

491

492

493

494

495

496

497

498

499

500

501

502

503

504

505

506

507

508

509

510

511

512

513

514

515

516

517

518

519

520

521

522

523

524

525

526

527

528

529

530

531

532

533

534

535

536

537

538

539

540

541

542

543

544

545

546

547

548

549

550

551

552

553

554

555

556

557

558

559

560

561

562

563

564

565

566

567

568

569

570

571

572

573

574

575

576

577

578

579

580

581

582

583

584

585

586

587

588

589

590

591

592

593

594

595

596

597

598

599

600

601

602

603

604

605

606

607

608

609

610

611

612

613

614

615

616

617

618

619

620

621

622

623

624

625

626

627

628

629

630

631

632

633

634

635

636

637

638

639

640

641

642

643

644

645

646

647

648

649

650

651

652

653

654

655

656

657

658

659

660

661

662

663

664

665

666

667

668

669

670

671

672

673

674

675

676

677

678

679

680

681

682

683

684

685

686

687

688

689

690

691

692

693

694

695

696

697

698

699

700

701

702

703

704

705

706

707

708

709

710

711

712

713

714

715

716

717

718

719

720

721

722

723

724

725

726

727

728

729

730

731

732

733

734

735

736

737

738

739

740

741

742

743

744

745

746

747

748

749

750

751

752

753

754

755

756

757

758

759

760

761

762

763

764

765

766

767

768

769

770

771

772

773

774

775

776

777

778

779

780

781

782

783

784

785

786

787

788

789

790

791

792

793

794

795

796

797

798

799

800

801

802

803

804

805

806

807

808

809

810

811

812

813

814

815

816

817

818

819

820

821

822

823

824

825

826

827

828

829

830

831

832

833

834

835

836

837

838

839

840

841

842

843

844

845

846

847

848

849

850

851

852

853

854

855

856

857

858

859

860

861

862

863

864

865

866

867

868

869

870

871

872

873

874

875

876

877

878

879

880

881

882

883

884

885

886

887

888

889

890

891

892

893

894

895

896

897

898

899

900

901

902

903

904

905

906

907

908

909

910

911

912

913

914

915

916

917

918

919

920

921

922

923

924

925

926

927

928

929

930

931

932

933

934

935

936

937

938

939

940

941

942

943

944

945

946

947

948

949

950

951

952

953

954

955

956

957

958

959

960

961

962

963

964

965

966

967

968

969

970

971

972

973

974

975

976

977

978

979

980

981

982

983

984

985

986

987

988

989

990

991

992

993

994

995

996

997

998

999

1000

1001

1002

1003

1004

1005

1006

1007

1008

1009

1010

1011

1012

1013

1014

1015

1016

1017

1018

1019

1020

1021

1022

1023

1024

1025

1026

1027

1028

1029

1030

1031

1032

1033

1034

1035

1036

1037

1038

1039

1040

1041

1042

1043

1044

1045

1046

1047

1048

1049

1050

1051

1052

1053

1054

1055

1056

1057

1058

1059

1060

1061

1062

1063

1064

1065

1066

1067

1068

1069

1070

1071

1072

1073

1074

1075

1076

1077

1078

1079

1080

1081

1082

1083

1084

1085

1086

1087

1088

1089

1090

1091

1092

1093

1094

1095

1096

1097

1098

1099

1100

1101

1102

1103

1104

1105

1106

1107

1108

1109

1110

1111

1112

1113

1114

1115

1116

1117

1118

1119

1120

1121

1122

1123

1124

1125

1126

1127

1128

1129

1130

1131

1132

1133

1134

1135

1136

1137

1138

1139

1140

1141

1142

1143

1144

1145

1146

1147

1148

1149

1150

1151

1152

1153

1154

1155

1156

1157

1158

1159

1160

1161

1162

1163

1164

1165

1166

1167

1168

1169

1170

1171

1172

1173

1174

1175

1176

1177

1178

1179

1180

1181

1182

1183

1184

1185

1186

1187

1188

1189

1190

1191

1192

1193

1194

1195

1196

1197

1198

1199

1200

1201

1202

1203

1204

1205

1206

1207

1208

1209

1210

1211

1212

1213

1214

1215

1216

1217

1218

1219

1220

1221

1222

1223

1224

1225

1226

1227

1228

1229

1230

1231

1232

1233

1234

1235

1236

1237

1238

1239

1240

1241

1242

1243

1244

1245

1246

1247

1248

1249

1250

1251

1252

1253

1254

1255

1256

1257

1258

1259

1260

1261

1262

1263

1264

1265

1266

1267

1268

1269

1270

1271

1272

1273

1274

1275

1276

1277

1278

1279

1280

1281

1282

1283

1284

1285

1286

1287

1288

1289

1290

1291

1292

1293

1294

1295

1296

1297

1298

1299

1300

1301

1302

1303

1304

1305

1306

1307

1308

1309

1310

1311

1312

1313

1314

1315

1316

1317

1318

1319

1320

1321

1322

1323

1324

1325

13

```
Applications Places Terminal Nov 7 18:09 student@ic312PD

File Edit View Search Terminal Help
Starting delay calculation for Hold views
AAE INFO: op1sDesignInPostRouteState() is 0
#####
# Design Stage: PreRoute
# Design Name: control_unit_fsm
# Design Mode: 50nm
# Analysis Mode: MMCM QCV
# Parasitics Mode: No SPEF/RCDB
# Signoff Settings: SI Off
#####
Start delay calculation (fullDC) (1 TI). (MEM=1700.9)
*** Calculating scaling factor for min timing libraries using the default operating condition of each library.
Total number of fetched objects 40
AAE INFO: Total number of nets for which stage creation was skipped for all views 0
End delay calculation. (MEM=1768.32 CPU=0:00:00.0 REAL=0:00:00.0)
End delay calculation (fullDC). (MEM=1768.32 CPU=0:00:00.0 REAL=0:00:00.0)
Turning on fast DC mode.
*** Done Building Timing Graph (cpu=0:00:00.1 real=0:00:00.0 totSessionCpu=0:01:17 mem=1776.3M)

-----
TimeDesign Summary
-----

Hold views included:
best_case

-----
+-----+-----+-----+
| Hold mode | all | reg2reg | default |
+-----+-----+-----+
| WNS (ns): | -0.173 | 0.003 | -0.173 |
| TNS (ns): | -0.703 | 0.000 | -0.703 |
| Violating Paths: | 5 | 0 | 5 |
| All Paths: | 12 | 7 | 5 |
+-----+-----+-----+

Density: 60.250%
Routing Overflow: 0.00% H and 0.00% V
-----
Reported timing to dir timingReports
Total CPU time: 0.19 sec
Total Real time: 1.0 sec
Total Memory Usage: 1694.59375 Mbytes
*** TimeDesign #3 [finish] : cpu/real = 0:00:00.2/0:00:00.2 (1.0), totSession cpu/real = 0:01:17.1/0:14:05.4 (0.1), mem = 1694.6M
Innovus 1>
Innovus 1> source ccopt.spec
Extracting original clock gating for clk...
Clock tree clk contains 4 sinks and 0 clock gates.
Extracting original clock gating for clk done.
The skew group clk/constraints pipeline was created. It contains 4 sinks and 1 sources.
Checking clock tree convergence...
Checking clock tree convergence done.
Innovus 2> 
```

```
Applications Places Terminal Nov 7 18:10 student@ic312PD

File Edit View Search Terminal Help
No ideal or dont touch nets found in the clock tree
No dont touch hnets found in the clock tree
No dont touch hpins found in the clock network.
Checking for illegal sizes of clock logic instances...
Checking for illegal sizes of clock logic instances done. (took cpu=0:00:00.0 real=0:00:00.0)
Validating CTS configuration done. (took cpu=0:00:00.0 real=0:00:00.0)

CCOpt configuration status: cannot run ccopt design.
Check the log for details of problem(s) found:

-----
Design configuration problems
-----
One or more clock trees have configuration problems
-----

Clock tree configuration problems:

-----
Clock tree Problem
-----
clk Could not determine drivers to use
-----

CCOpt::Phase::PreparingToBalance done. (took cpu=0:00:00.1 real=0:00:00.1)
Runtime done. (took cpu=0:00:00.1 real=0:00:00.1)
Runtime Summary
=====
Clock Runtime: (71%) Core CTS 0.05 (Init 0.05, Construction 0.00, Implementation 0.00, eORPC 0.00, PostConditioning 0.00, Other 0.00)
Clock Runtime: (0%) CTS services 0.00 (RefinePlace 0.00, EarlyGlobalClock 0.00, NanoRoute 0.00, ExtractRC 0.00, TimingAnalysis 0.00)
Clock Runtime: (28%) Other CTS 0.02 (Init 0.02, CongRepair/EGR-DP 0.00, TimingUpdate 0.00, Other 0.00)
Clock Runtime: (100%) Total 0.06

Leaving CCOpt scope - Cleaning up placement interface...
Leaving CCOpt scope - Cleaning up placement interface done. (took cpu=0:00:00.0 real=0:00:00.0)
***ERROR: 3
*** CTS #1 [finish] (ccopt design #1) : cpu/real = 0:00:00.1/0:00:00.1 (0.9), totSession cpu/real = 0:01:25.2/0:16:07.5 (0.1), mem = 1702.9M
***ERROR: (IMPCCOPT-2196): Cannot run ccopt design because the command prerequisites were not met. Review the previous error messages for more details about the failure.

*** Summary of all messages that are not suppressed in this session:
Severity ID Count Summary
ERROR IMPCCOPT-1135 1 CTS found neither inverters nor buffers ...
ERROR IMPCCOPT-2196 1 Cannot run ccopt design because the comm...
WARNING IMPCCOPT-1183 1 The library has no usable balanced %ss f...
WARNING IMPCCOPT-1184 1 The library has no usable balanced %ss f...
*** Message Summary: 2 warning(s), 2 error(s)

*** ccopt design #1 [finish] : cpu/real = 0:00:00.1/0:00:00.1 (1.0), totSession cpu/real = 0:01:25.2/0:16:07.5 (0.1), mem = 1702.9M
#% End ccopt design (date=11/07 18:10:14, total cpu=0:00:00.1, real=0:00:00.0, peak res=1689.0M, current mem=1689.0M)

Innovus 3> 
```

```
Applications Places Terminal Nov 7 18:10 student@ic312PD

File Edit View Search Terminal Help
No ideal or dont_touch nets found in the clock tree
No dont_touch hnets found in the clock tree
No dont_touch hpins found in the clock network.
Checking for illegal sizes of clock logic instances...
Checking for illegal sizes of clock logic instances done. (took cpu=0:00:00.0 real=0:00:00.0)
Validating CTS configuration done. (took cpu=0:00:00.0 real=0:00:00.0)

CCOpt configuration status: cannot run ccopt design.
Check the log for details of problem(s) found:

-----
Design configuration problems
-----
One or more clock trees have configuration problems
-----

Clock tree configuration problems:
-----
Clock tree Problem
-----
clk Could not determine drivers to use
-----

CCOpt::Phase::PreparingToBalance done. (took cpu=0:00:00.1 real=0:00:00.1)
Runtime done. (took cpu=0:00:00.1 real=0:00:00.1)
Runtime Summary
=====
Clock Runtime: (71%) Core CTS 0.05 (Init 0.05, Construction 0.00, Implementation 0.00, eGRPC 0.00, PostConditioning 0.00, Other 0.00)
Clock Runtime: (0%) CTS services 0.00 (RefinePlace 0.00, EarlyGlobalClock 0.00, NanoRoute 0.00, ExtractRC 0.00, TimingAnalysis 0.00)
Clock Runtime: (28%) Other CTS 0.02 (Init 0.02, CongRepair/EGR-DP 0.00, TimingUpdate 0.00, Other 0.00)
Clock Runtime: (100%) Total 0.06

Leaving CCOpt scope - Cleaning up placement interface...
Leaving CCOpt scope - Cleaning up placement interface done. (took cpu=0:00:00.0 real=0:00:00.0)
**ERROR: 3
*** CTS #1 [finish] (ccopt design #1) : cpu/real = 0:00:00.1/0:00:00.1 (0.9), totSession cpu/real = 0:01:25.2/0:16:07.5 (0.1), mem = 1702.9M
**ERROR: (IMPCCOPT-2196): Cannot run ccopt design because the command prerequisites were not met. Review the previous error messages for more details about the failure.

*** Summary of all messages that are not suppressed in this session:
Severity ID Count Summary
ERROR IMPCCOPT-1135 1 CTS found neither inverters nor buffers ...
ERROR IMPCCOPT-2196 1 Cannot run ccopt design because the comm...
WARNING IMPCCOPT-1183 1 The library has no usable balanced %ss f...
WARNING IMPCCOPT-1184 1 The library has no usable balanced %ss f...
*** Message Summary: 2 warning(s), 2 error(s)

*** ccopt design #1 [finish] : cpu/real = 0:00:00.1/0:00:00.1 (1.0), totSession cpu/real = 0:01:25.2/0:16:07.5 (0.1), mem = 1702.9M
%# End ccopt design (date=11/07 18:10:14, total cpu=0:00:00.1, real=0:00:00.0, peak res=1689.0M, current mem=1689.0M)

innovus > saveDesign DBS/cts.enc3
```

```
Applications Places Terminal Nov 7 18:10 student@ic312PD

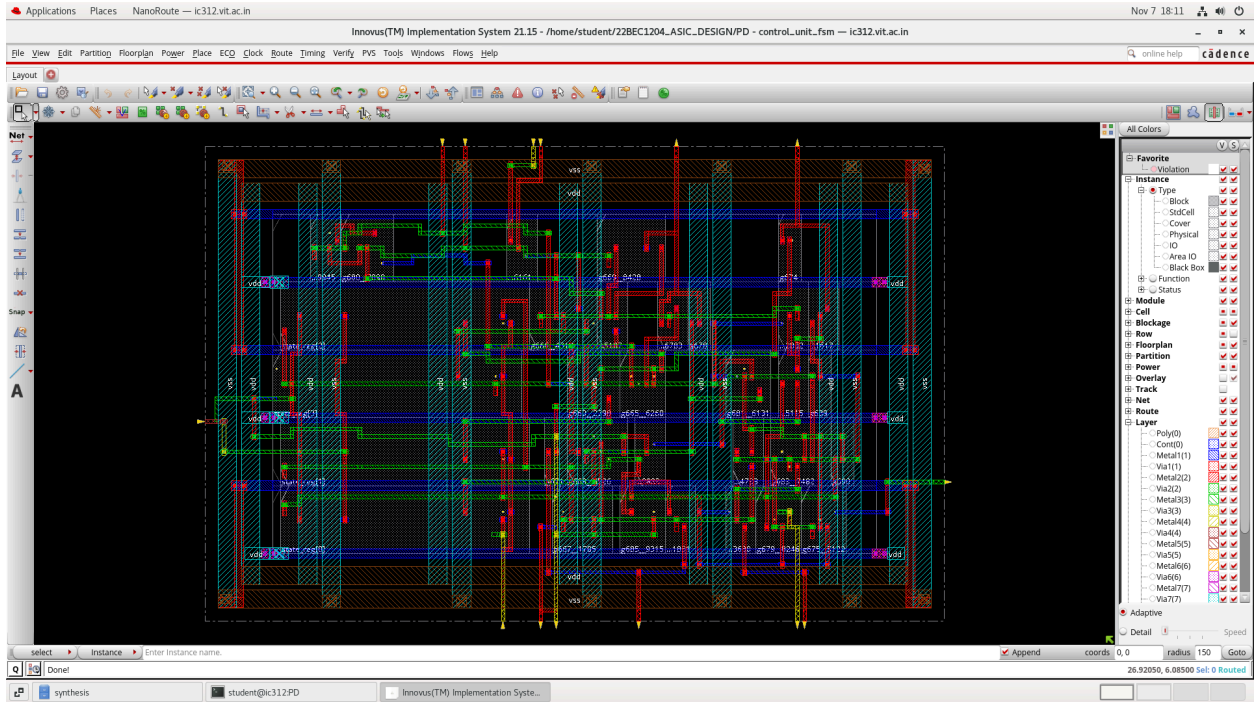
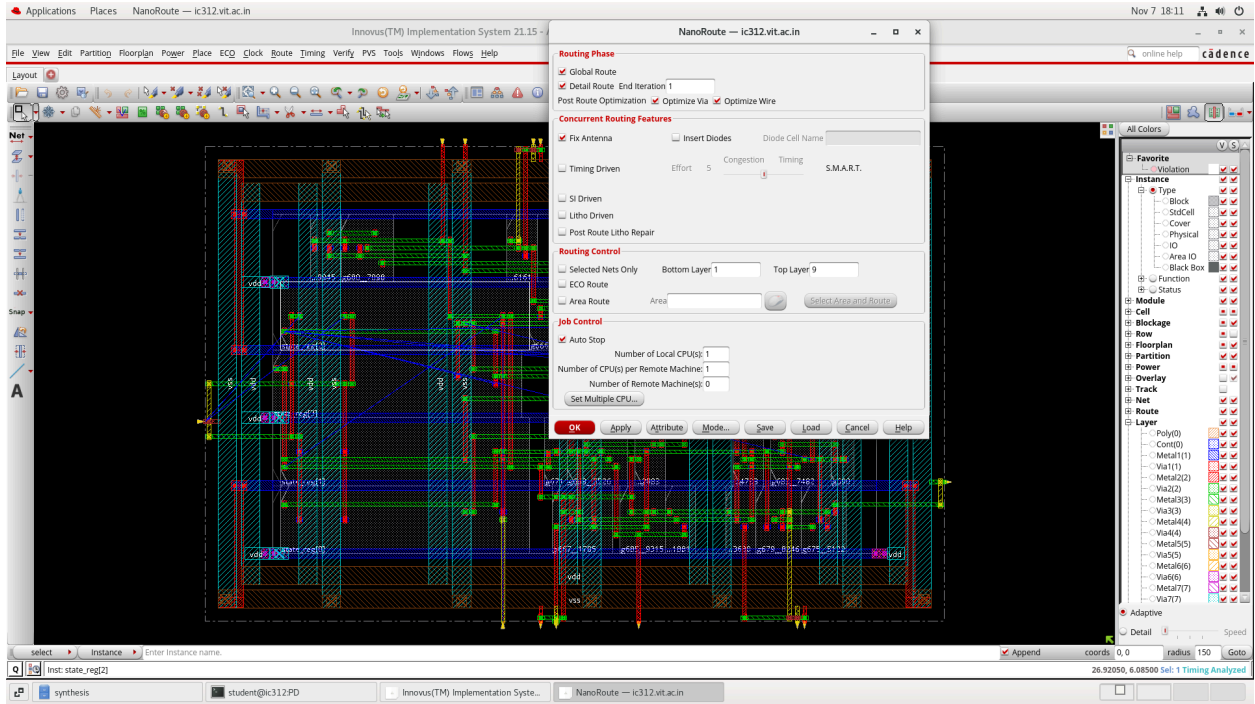
File Edit View Search Terminal Help
*** CTS #1 [finish] (ccopt design #1) : cpu/real = 0:00:00.1/0:00:00.1 (0.9), totSession cpu/real = 0:01:25.2/0:16:07.5 (0.1), mem = 1702.9M
**ERROR: (IMPCCOPT-2196): Cannot run ccopt design because the command prerequisites were not met. Review the previous error messages for more details about the failure.

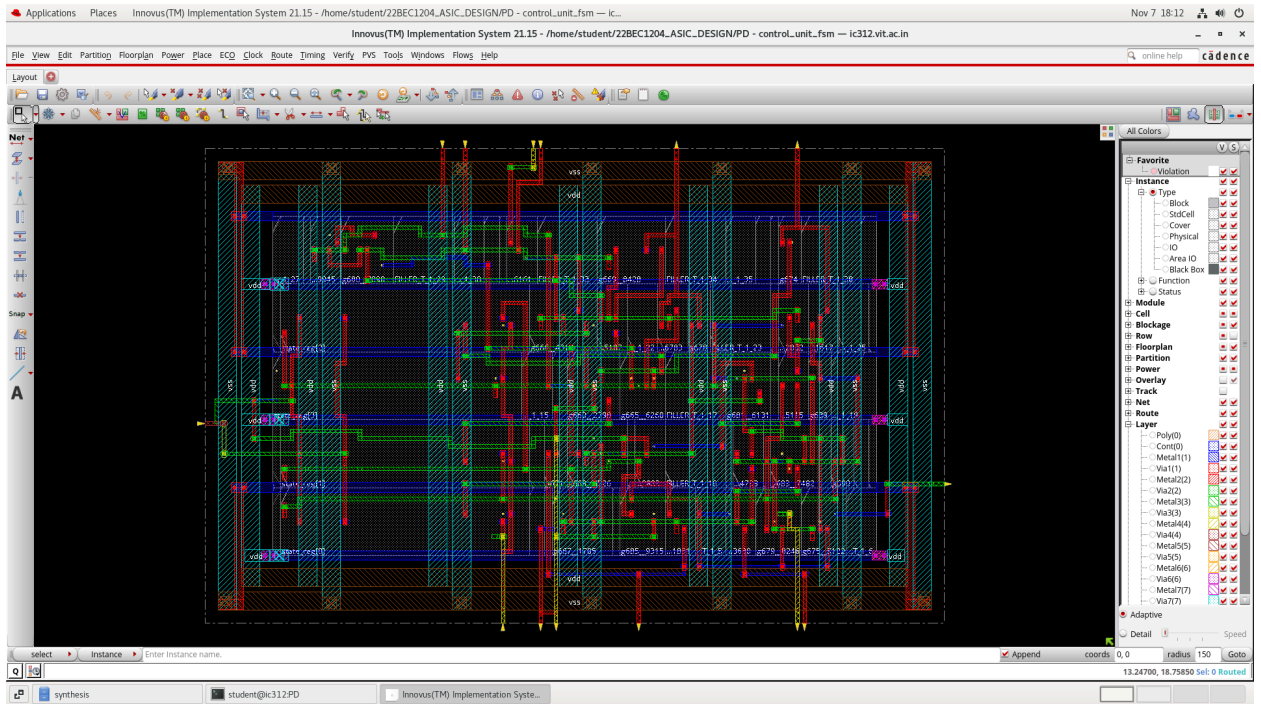
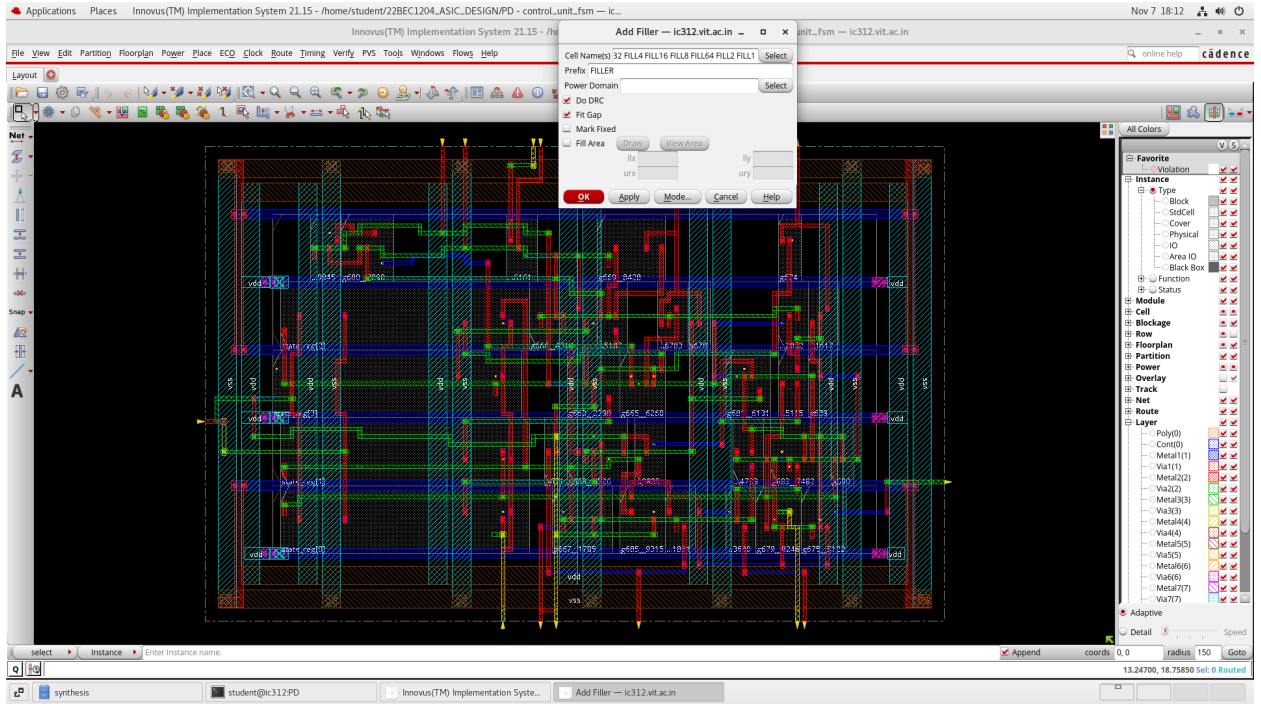
*** Summary of all messages that are not suppressed in this session:
Severity ID Count Summary
ERROR IMPCCOPT-1135 1 CTS found neither inverters nor buffers ...
ERROR IMPCCOPT-2196 1 Cannot run ccopt design because the comm...
WARNING IMPCCOPT-1183 1 The library has no usable balanced %ss f...
WARNING IMPCCOPT-1184 1 The library has no usable balanced %ss f...
*** Message Summary: 2 warning(s), 2 error(s)

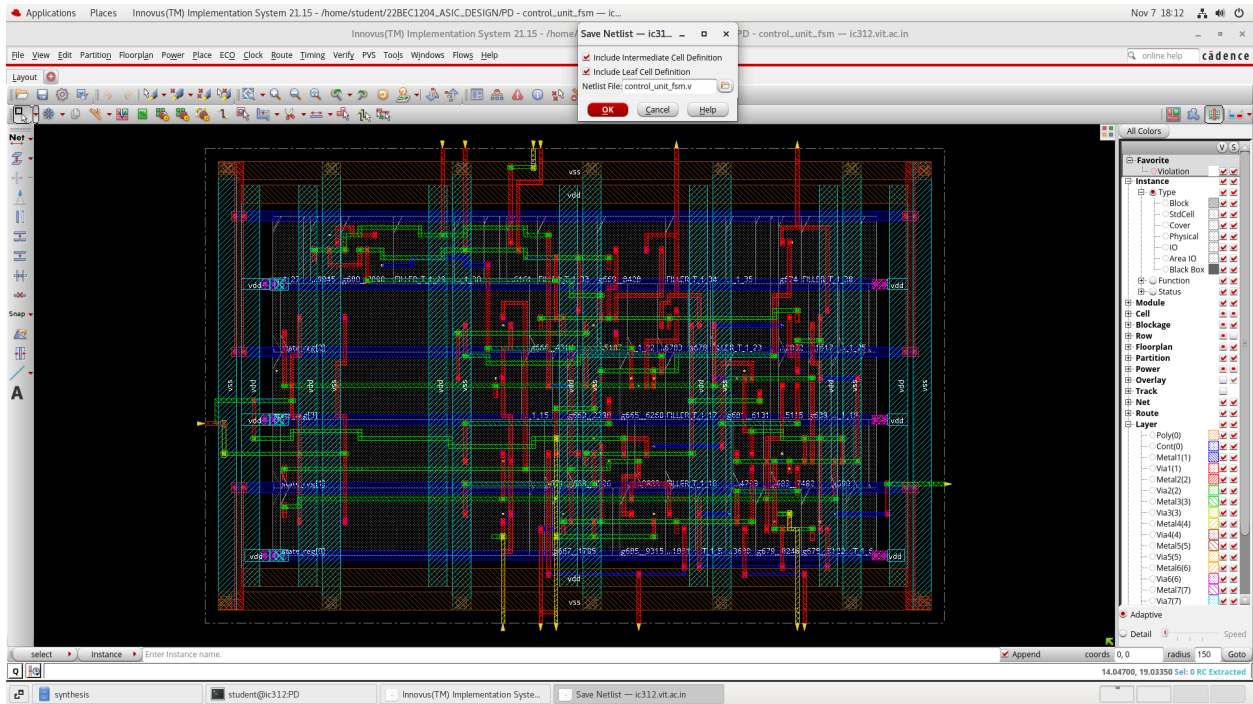
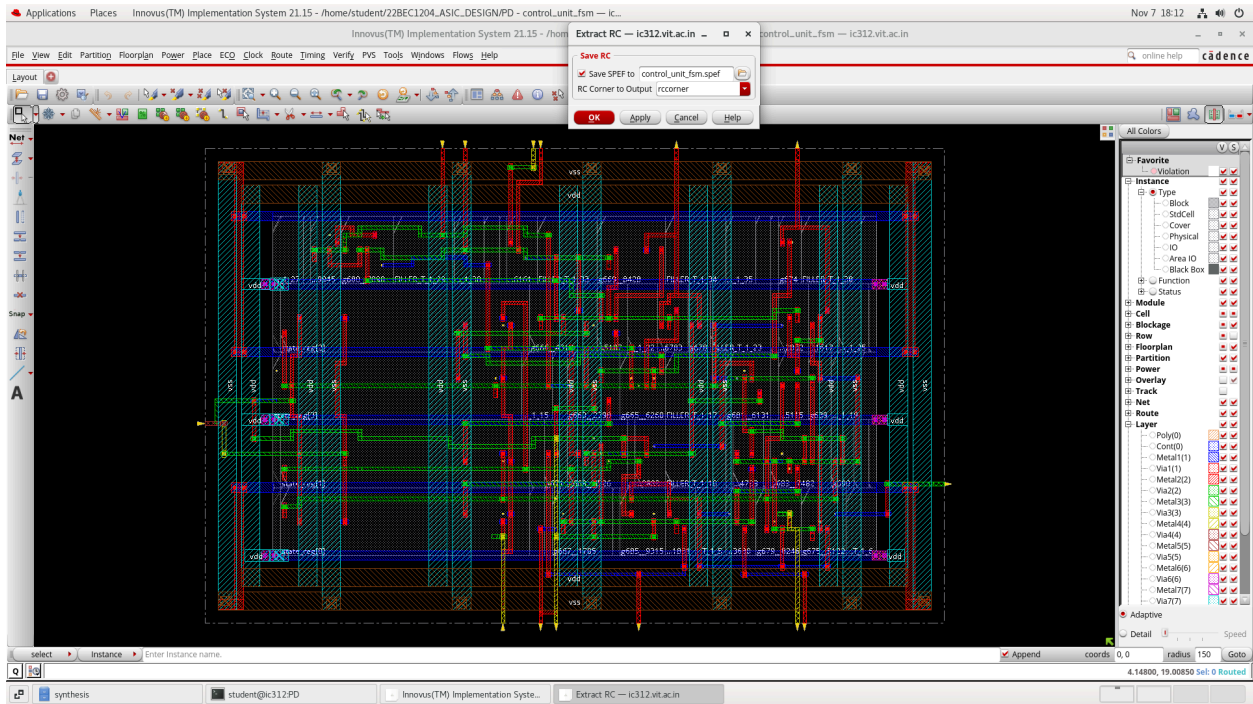
*** ccopt design #1 [finish] : cpu/real = 0:00:00.1/0:00:00.1 (1.0), totSession cpu/real = 0:01:25.2/0:16:07.5 (0.1), mem = 1702.9M
%# End ccopt design (date=11/07 18:10:14, total cpu=0:00:00.1, real=0:00:00.0, peak res=1689.0M, current mem=1689.0M)

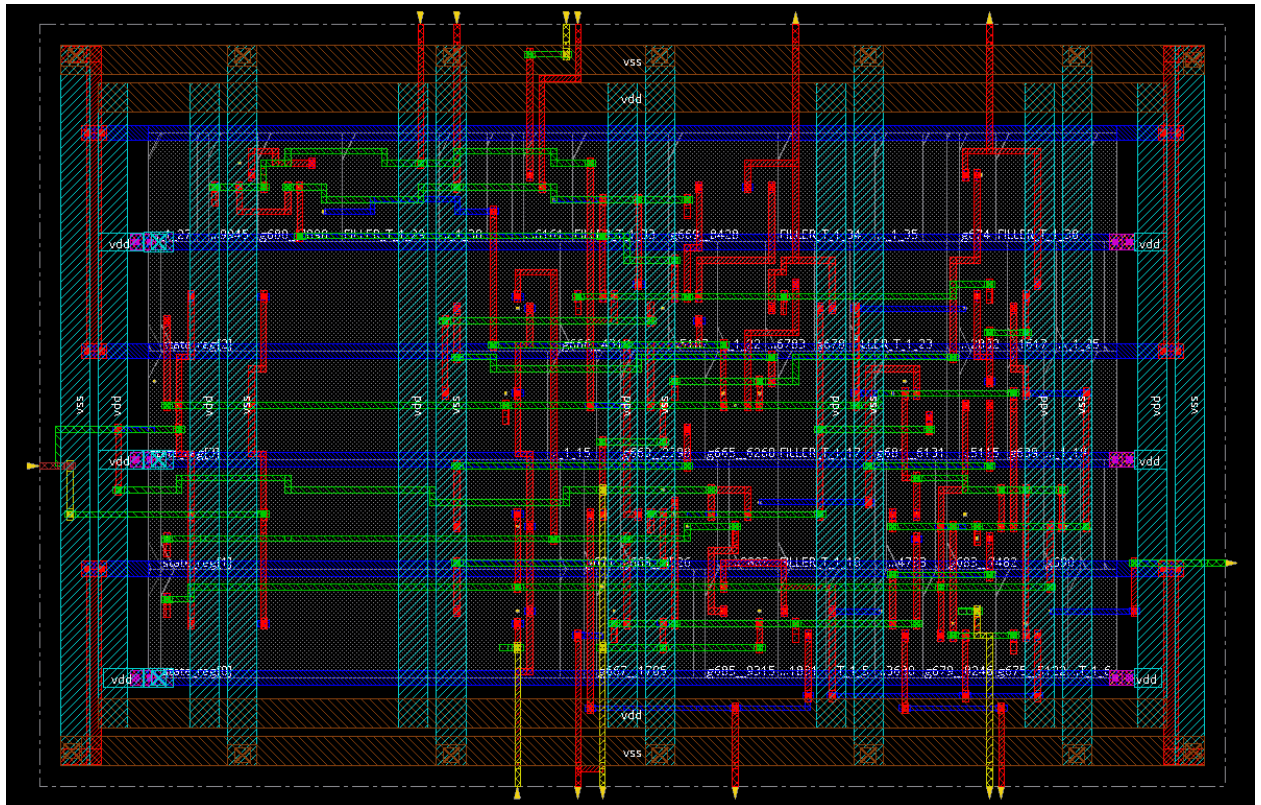
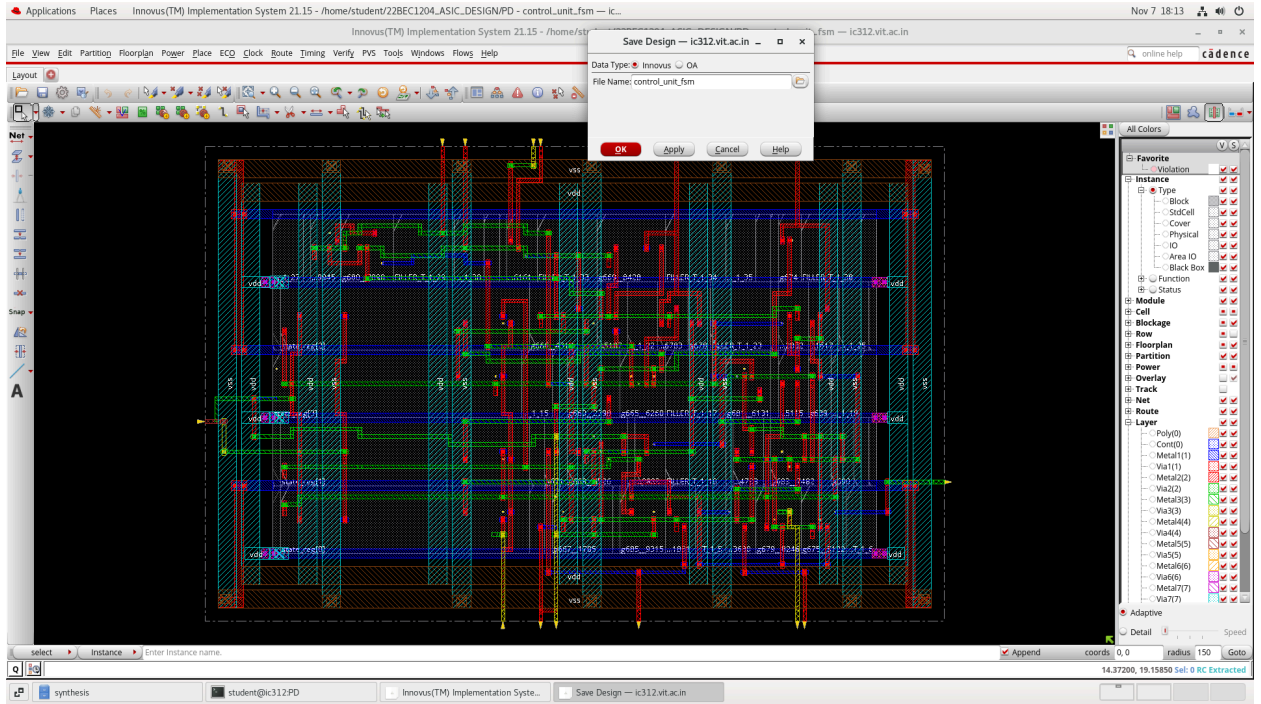
innovus > saveDesign DBS/cts.enc3
%# Begin save design ... (date=11/07 18:10:53, mem=1694.5M)
%# Begin Save ccopt configuration ... (date=11/07 18:10:53, mem=1694.5M)
%# End Save ccopt configuration ... (date=11/07 18:10:53, total cpu=0:00:00.0, real=0:00:00.0, peak res=1695.1M, current mem=1695.1M)
%# Begin Save netlist data ... (date=11/07 18:10:53, mem=1695.1M)
%# End Save netlist data ... (date=11/07 18:10:53, total cpu=0:00:00.0, real=0:00:00.0, peak res=1695.3M, current mem=1695.3M)
Writing Binary DB to DBS/cts.enc3.dat/control unit fsm.v.bin in single-threaded mode...
%# End Save netlist data ... (date=11/07 18:10:53, total cpu=0:00:00.0, real=0:00:00.0, peak res=1695.3M, current mem=1695.3M)
Saving symbol table file ...
Saving congestion map file DBS/cts.enc3.dat/control unit fsm.route.congmap.q2 ...
%# Begin Save AAE data ... (date=11/07 18:10:53, mem=1695.6M)
Saving AAE Data ...
%# End Save AAE data ... (date=11/07 18:10:53, total cpu=0:00:00.0, real=0:00:00.0, peak res=1695.6M, current mem=1695.6M)
Saving preference file DBS/cts.enc3.dat/gui.pref.tcl ...
Saving mode setting ...
Saving global file ...
%# Begin Save floorplan data ... (date=11/07 18:10:53, mem=1697.5M)
Saving floorplan file ...
%# End Save floorplan data ... (date=11/07 18:10:53, total cpu=0:00:00.0, real=0:00:00.0, peak res=1697.6M, current mem=1697.6M)
Saving Drc markers ...
... No Drc file written since there is no markers found.
%# Begin Save placement data ... (date=11/07 18:10:53, mem=1697.6M)
%# Saving stdCellPlacement binary (version 2) ...
Save Adaptive View Pruning View Names to Binary file
%# End Save placement data ... (date=11/07 18:10:53, total cpu=0:00:00.0, real=0:00:00.0, peak res=1697.9M, current mem=1697.9M)
%# Begin Save routing data ... (date=11/07 18:10:53, mem=1697.9M)
Saving route file ...
*** Completed saveRoute (cpu=0:00:00.0 real=0:00:01.0 mem=1705.7M) ***
%# End Save routing data ... (date=11/07 18:10:54, total cpu=0:00:00.0, real=0:00:01.0, peak res=1698.2M, current mem=1698.2M)
Saving property file DBS/cts.enc3.dat/control unit fsm.prop
*** Completed saveProperty (cpu=0:00:00.0 real=0:00:00.0 mem=1708.7M) ***
%# Begin Save power constraints data ... (date=11/07 18:10:54, mem=1699.4M)
%# End Save power constraints data ... (date=11/07 18:10:54, total cpu=0:00:00.0, real=0:00:00.0, peak res=1699.4M, current mem=1699.4M)
Generated self-contained design cts.enc3.dat
%# End save design ... (date=11/07 18:10:54, total cpu=0:00:00.2, real=0:00:01.0, peak res=1728.2M, current mem=1702.7M)
*** Message Summary: 0 warning(s), 0 error(s)

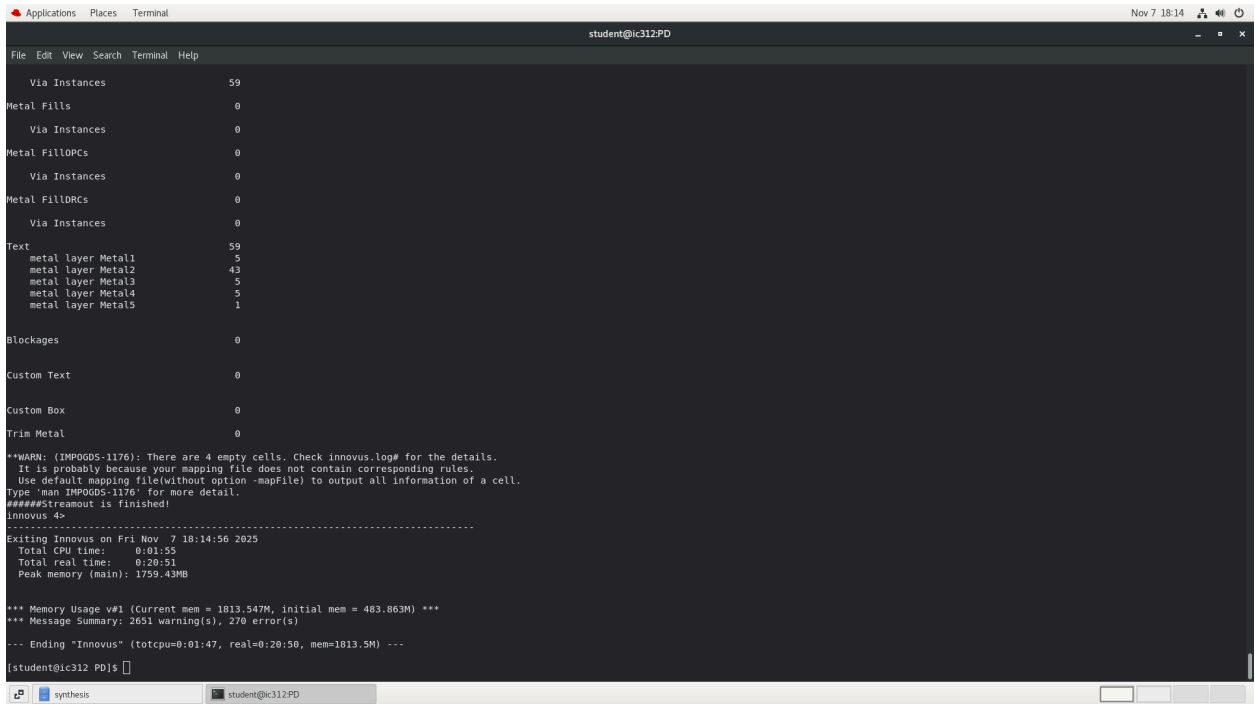
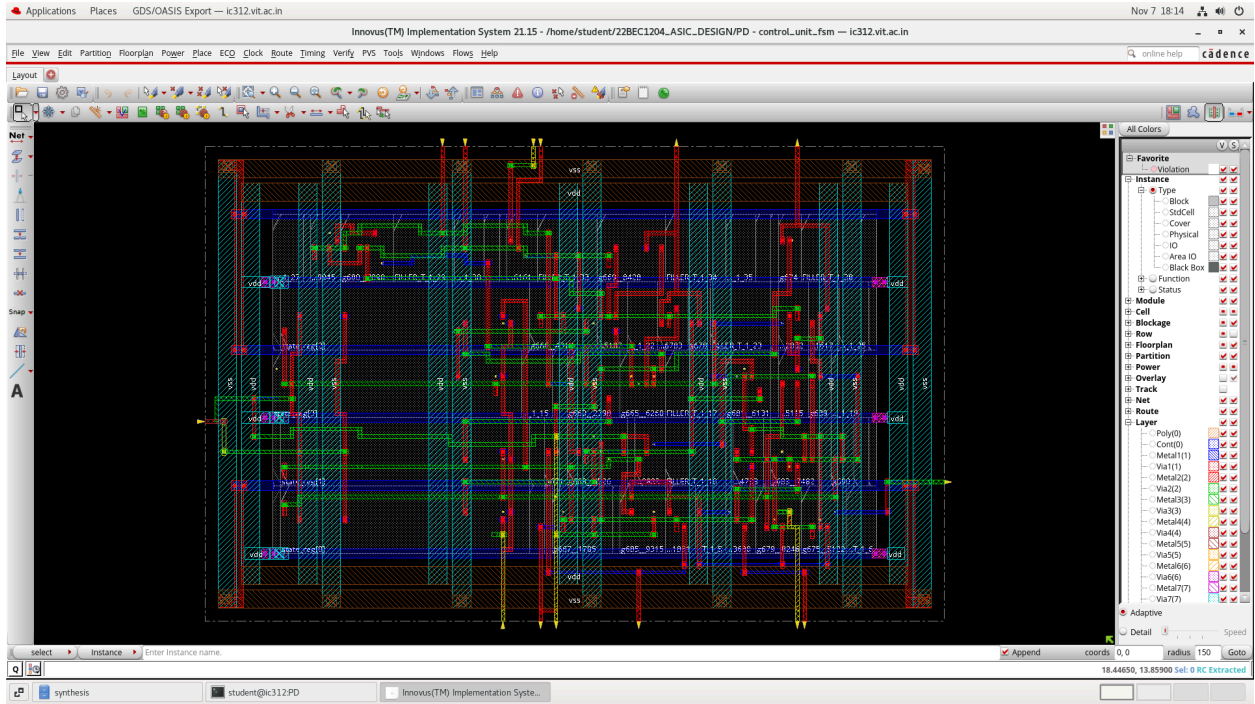
0
innovus >
```











7. Inference

A single-cycle FSM control unit was successfully designed, verified, and implemented using a complete ASIC flow. The design correctly implements a Moore FSM by separating sequential state registers from the combinational next-state and output logic.

The design passed all stages, from functional simulation to a final, routed GDSII file. The successful DFT insertion and ATPG run confirmed the testability of the design, which was possible because the FSM's state registers provided the necessary scannable flops. This project demonstrates a successful implementation of a core sequential component of a CPU.